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# Exploring Jet Substructure in Public LHC Data

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## AGRADECIMENTOS

# RESUMO

**Palavras-chave:**

# ABSTRACT

**Keywords:**

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# LIST OF ABBREVIATIONS AND SYMBOLS

# 1 INTRODUCTION

## 1.1 Context and motivation

The Large Hadron Collider (LHC) at CERN brings the highest energy proton-proton collisions ever conceived in a laboratory and, with them, an enourmous sample of strongly interacting final states. The majority of these collisions are governed by the Quantum Chromodynamics (QCD), the gauge theory of the strong interaction. The comprehension and study of these collisions are not only a prerequisite for surpressing backgrounds in searches for new phenomena but also the structure of the hadronic final state can be examined and compared to theory directly, at scales where the coupling is weak enough for peturbation theory to be applied and yet the dynamics still remain rich.

The most prominent objects in these final states are jets. Jets are collimated sprays of hadrons that emerge from the fragmentation of high energy quarks and gluons. A jet is not a fundamental object but instead a definition. A clustering algorithm specifies which detector inputs are grouped together and a recombination scheme specifies how momentum is assigned to a result. This defines a jet, from which an interesting and important question then arises. What does the radiation inside the jets look like? This domain is called jet substructure, which over the past decade has grown from a collection of tagging tools into a precision probe of QCD dynamics in its own right.

In order to organise and look at the internal structure of a jet, one can use the Lund jet plane [1]. The Lund jet plane is a two dimensional map of a jet's branching history on which soft, collinear and non perturbative regions of QCD radiation occupy distinct territories. This map records where the jet's internal emissions occur, but it sets aside an intriguing piece of information, the orientation of each splitting relative to the one before. This orientation is not random. Through the linear polarisation of intermediate gluons, successive splittings are correlated in their relative azimuthal angle, producing a characteristic modulation in the distribution of that angle. This is a genuine quantum interference effect which survives only because the intermediate gluon is described by an amplitude rather than a probability, and it is absent from any picture that treats successive emissions as independent classical branchings.

These collinear spin correlations are well understood theoretically and are implemented in modern parton shower algorithms, yet establishing them cleanly in collision data has proved rather difficult. The reason is that the effect carries opposite signs in different splitting channels. The $g \to q\bar{q}$ and $g \to gg$ channels contribute with opposite sign, so a measurement that sums all channels sees the two contributions partially cancel. This interference is not absent from such an inclusive measurement. It is diluted and the apparent size reflects the accidental flavour composition of the sample rather than the strength of the underlying physics.

This is the obstacle the present work addresses. A meaningful measurement of the effect requires knowing which splitting produced each emission. However, the flavour of an individual splitting inside a jet is not directly observable and must be inferred. Working entirely on public CERN Open Data [2], more precisely with CMS Open Data [3], this dissertation builds a reproducible pipeline from the released samples through Lund declustering to the azimuthal spin observable, and introduces a flavour selection that isolates the $g \to q\bar{q}$ channel. The resulting modulation is reported not as a single number but as a function of channel purity achieved by that selection, a presentation that separates the physics of the

interference from the performance of the selection and allows the magnitude of the effect to be read off in the limit of a pure sample.

The Lund declustering angle $\Delta\psi_{12}$ is not the only way to reach this effect. The same $\cos 2\psi$ modulation also imprints itself on the flow of energy within a jet, and can be accessed through a three-point energy correlator, which probes the relative azimuthal orientation of energy deposited by triplets of particles without reference to any clustering history. The two observables are complementary: one reads the effect from the reconstructed branching sequence, the other from the energy flow directly, and each carries a different exposure to the choices and imperfections of the measurement. This work develops both, treating the Lund observable as the primary measurement and the energy correlator as an independent cross-check on the same underlying physics.

The fact that this analysis rests on public data is not incidental. The CERN Open Data Portal [2] has made it possible for work outside the experimental collaborations to produce meaningful results and to carry out genuinely new measurements. The present work is intended as a contribution in that spirit. It builds an end to end pipeline from a single CMS dataset to the final measurements, establishing along the way a workflow for accessing, handling and measuring with CMS Open Data.

## 1.2 Objectives

The main aim of this dissertation is to measure collinear spin correlations in QCD jets using public LHC data and to establish the flavour selection that such a measurement requires in order to be meaningful. For the purpose of achieving that goal, the work is organised around a sequence of objectives. The first is to build a reproducible pipeline over CERN Open Data samples capable of extracting the necessary variables to begin a jet substructure analysis. This means that the data samples must contain jet constituent information, thus the data format chosen must retain that level of detail, ruling out the more compact samples in favour of one that preserves per-constituent quantities.

With that pipeline in place, the second objective is to construct the primary and secondary Lund jet planes and to verify that they reproduce the known features of QCD radiation, since this agreement is what establishes that the declustering is sound before any finer observable is built on top of it. Following that, the third objective is to implement the azimuthal observable $\Delta\psi_{12}$ between the hardest primary and secondary branchings, and to study its distribution both with and without grooming, recovering the inclusive result before attempting to sharpen it.

The fourth objective is the central one, and the point at which this work departs from a purely inclusive analysis. Because the spin-interference signal carries opposite signs in the $g \to q\bar{q}$ and $g \to gg$ channels, an inclusive measurement dilutes it. Isolating a single channel is therefore not a refinement but a precondition for measuring the effect at its true strength. This objective is thus to develop a flavour selection capable of enriching the $g \to q\bar{q}$ channel, and to quantify both how well it performs and how pure a sample it can reach in principle. The fifth objective builds directly on it. It is to measure $\langle \cos 2\psi_{12} \rangle$ as a function of the channel purity delivered by that selection and to establish the value characteristic of a pure $g \to q\bar{q}$ sample.

Finally, the work aims to assess the systematic uncertainties affecting this measurement and to identify the dominant limitations of the approach, so that the extensions it opens up can be set out on solid ground.

## 1.3 Thesis outline

Chapter 2 develops the theoretical material on which the measurement rests, such as the elements of QCD relevant to jet formation, the definition and reconstruction of jets and the algorithms used to build and decluster them, the Lund jet plane, and the origin of collinear spin correlations together with the two observables through which this work accesses them. Chapter 3 turns to the data and the analysis, and covers the CERN Open Data resources and the CMS detector, the data format and dataset used, and the pipeline that runs from the released samples through Lund declustering to the azimuthal observable, the flavour selection, and the treatment of systematic uncertainties. Chapter 4 presents the validation of the pipeline and the measurement itself, and discusses the results in the light of the expectations set out in Chapter 2. Finally, Chapter 5 returns to the objectives of Section 1.2, states what has been established and with what confidence, and outlines the extensions that the present framework makes possible.

# 2 THEORETICAL BACKGROUND

## 2.1 Quantum Chromodynamics

Quantum Chromodynamics (QCD) is the quantum field theory of the strong interaction, a non-Abelian gauge theory built on the colour symmetry group $SU(3)$ [4]. Its fundamental degrees of freedom are quarks, which carry colour charge, and gluons, the gauge bosons that mediate the interaction. The non-Abelian character of the theory has a decisive consequence absent from Quantum Electrodynamics. The gluons themselves carry colour, and therefore couple directly to one another. To a large extent, the dynamics of the strong interaction are the dynamics of this gluon self-coupling, and the two properties of QCD that govern everything in this chapter both follow from it.

### 2.1.1 Asymptotic freedom and confinement

The first property is asymptotic freedom. The strong coupling $\alpha_s$ decreases as the momentum transfer of an interaction increases, so that at the high scales probed in LHC collisions the quarks and gluons interact weakly and perturbation theory applies [5, 6]. The running of the coupling is captured at leading order by

$$\alpha_s(Q^2) = \frac{\alpha_s(\mu^2)}{1 + \alpha_s(\mu^2)\,\frac{33 - 2n_f}{12\pi}\,\ln\frac{Q^2}{\mu^2}}, \tag{2.1}$$

where $Q$ is the scale of the process, $\mu$ a reference scale, and $n_f$ the number of active quark flavours. The positive sign of the coefficient $33 - 2n_f$, which holds for any physical number of flavours, is what makes the coupling fall at large $Q$. It is a direct imprint of the gluon self-interaction, and its discovery established that a perturbative treatment of hard processes is justified. It is convenient to trade the reference scale for a single dimensionful constant $\Lambda_{\text{QCD}}$, the scale at which the perturbative coupling formally diverges, of an order of a few hundred MeV. In terms of it the coupling is small for $Q \gg \Lambda_{\text{QCD}}$ and grows strong as $Q$ approaches it.

This growth is the second property called confinement. At low momentum transfer $\alpha_s$ becomes large and colour-charged particles cannot be isolated. A quark or gluon produced in a hard scattering does not propagate freely to a detector, because as it separates from other colour charges, the energy stored in the colour field between them grows with distance until it becomes energetically favourable to create a new quark-antiquark pairs from the vacuum. The original parton is thereby dressed into a spray of colour-neutral hadrons. This is why the strongly interacting final state reaches the detector not as partons but as the collimated sprays introduced in Chapter 1, and why any comparison between a perturbative calculation and a measurement must bridge the divide between the partonic and hadronic descriptions.

### 2.1.2 Soft and collinear radiation

The bridge between the hard parton and the observed hadrons is built from successive emissions, and the pattern of those emissions is far from uniform. When a parton radiates a gluon, the emission probability

at first order in the coupling takes the schematic form

$$d\mathcal{P} \propto \frac{2\alpha_s C}{\pi} \frac{d\theta}{\theta} \frac{dE}{E}, \tag{2.2}$$

where $E$ and $\theta$ are the energy and emission angle of the radiated gluon and $C$ is a colour factor. The two factors carry the characteristic structure of QCD radiation. The factor $d\theta/\theta$ diverges as $\theta \to 0$ meaning that emissions are enhanced when the gluon is radiated close to the direction of its emmiter. This is known as the collinear divergence. The factor $dE/E$ diverges as $E \to 0$ meaning that emissions are enhanced when the gluon is soft. This is known as the soft divergence. Together they mean that a hard parton radiates predominantly softly and at small angles, and that number of such emissions is, in a precise logarithmic sense, large.

The collinear structure is described more completely by the Dokshitzer–Gribov–Lipatov–Altarelli–Parisi (DGLAP) splitting functions [4], which give the probability that a parton of one type splits into two, as a function of the momentum fraction $z$ carried by one of the products. For the three splittings relevant to this work they read, at leading order,

$$P_{q\to qg}(z) = C_F \frac{1+z^2}{1-z}, \tag{2.3}$$

$$P_{g\to gg}(z) = C_A \left[\frac{z}{1-z} + \frac{1-z}{z} + z(1-z)\right], \tag{2.4}$$

$$P_{g\to q\bar{q}}(z) = T_R \left[z^2 + (1-z)^2\right], \tag{2.5}$$

with colour factors $C_F = 4/3$, $C_A = 3$ and $T_R = 1/2$. The distinction between the gluon splitting channels of Eqs. (2.4) and (2.5), one producing a pair of gluons and the other a quark-antiquark pair, will prove central. The two channels differ not only in these rates but in the spin structure of their emissions, and it is that difference which the spin observables of Section 2.4 are designed to probe.

These divergences are not a pathology. They are integrable, and in any physical, infrared-safe observable the divergent real emissions cancel against corresponding virtual corrections, leaving a finite result. This cancellation is the requirement that shapes the very definition of a jet in Section 2.2. Their logarithmic structure, however, is physical and measurable, and it is exactly the structure that the Lund jet plane of Section 2.3 is built to expose. By mapping each emission onto logarithmic axes of angle and transverse momentum, it turns the double logarithmic enhancement of equation (2.2) into an approximately uniform density, against which finer correlations stand out.

### 2.1.3 Parton showers and hadronization

Because a single hard parton typically radiates many times before hadronizing, the emissions can be organised into a sequence orderred from the hardest, widest-angle splittings down to the soft and collinear ones. This sequence is what is known as a parton shower. Modern Monte Carlo event generators implement this sequence probabilistically through the splitting functions of equations (2.3)-(2.5), evolving each parton down from the hard scale of the collision to a low scale of order $1$ GeV, at which the perturbative description breaks down and confinement takes over. Below that scale the partons are converted into hadrons by phenomenological hadronization models, tuned to data rather than derived from first principles.

This picture, a hard, perturbatively calculable core, dressed by a shower of increasingly soft and collinear emissions and finally hadronized, underlies the whole of the analysis that follows. The observables studied in this work are built to read the perturbative shower through the hadrons that reach the detector, and for this reason the measurement is deliberately confined to a hard kinematic regime, where the perturbative component dominates and non-perturbative corrections are comparatively suppressed.

## 2.2 Jets

As settled in Section 2.1, a coloured parton produced in a hard scattering does not reach the detector directly because it radiates and hadronizes before arriving as a collimated spray of hadrons. A jet is then an object, which by grouping those hadrons and combining their momentum back together inverts the process approximately and estimates the momentum of the orininal parton that produced them. Therefore, a jet is a reconstructed physical object, built from a construction algorithm, with properties that depend on this specific algorithm used to build it. This section sets out that the requirement which any jet definition must satisfy, in order to be usable in peturbative QCD, and three algorithms, used in this work and built on that requirement.

### 2.2.1 Jet algorithms and infrared–collinear safety

A jet is built using two components. A jet algorithm establishes which inputs, from detector-level energy deposits, reconstructed particles, or, in a calculation, partons, are grouped together into a jet. A recombination scheme picks up those grouped inputs and specifies how their momenta is combined in order to compute the momentum of the jet. Across this work, the inputs are the particle-flow candidates, described in Chapter 3, and momenta are combined in the $E$-scheme, in which four-momenta are simply added. This scheme is Lorentz invariant and returns jets with non-zero mass.

Though, every conceivable grouping prescription is not admissible. The requirement necessary and the one that eliminates most of them is infrared and collinear (IRC) safety. In order for an observable to be infrared safe is that its value most remain unchanged by the addition of an arbitrarily soft emission. For it to be collinear safe, this value most also reamin unchanged if one of its inputs is replaced by two collinear inputs sharing its momentum. So, a jet algorithm is IRC safe when the set of jets it returns is stable under both of these operations. A soft particle must not create or destroy a jet and a collinear splitting must not change how particles are assigned to jets.

This requirement is not optional because of what follows directly from Section 2.1. The emission probability of equation (2.2) diverges in both the soft and the collinear limits. In any IRC-safe observable these divergent real emissions cancel against equally divergent virtual corrections, leaving a finite result. Opposingly, in an IRC-unsafe observable, that cancellation is ruined and a fixed-order peturbative prediction returns infinity rather than an finite number. An IRC-unsafe jet definition is therefore not merely inconvenient but it also cannot be compared with perturbative QCD at all. Since the purpose of measuring jets is to confront the theory with data, IRC safety is a precondition and not a refinement.

In the historical development of jet algorithms this criteria has been conclusive. The first definitions that appeared were cone algorithms, which grouped particles falling within a fixed angular radius of a candidate axis. These algorithms were simple and seed-based and were not IRC safe because a soft particle could seed a new candidate cone, or a collinear splitting could change which particles acted as

seeds, thus either these flaws could alter the resulting jets. A lot of effort was put in order to fix them and eventually producing seedless variants that restored collinear safety. Alternatively, another family of algorithms, the sequential recombination algorithms, are IRC safe by construction and this work utilised them.

Since the same mechanism reappears when flavour is assigned to jets in Section 2.2.4, it is worh making concrete how such a problem arises. Consider a seed-based cone algorithm applied to an event with two hard particles whose separation is slightly greater than the cone radius, so that each seeds its own jet. By adding a single soft particle between them, it can seed a third cone that overlaps both, merging the two hard particles into one jet where before there were only two, which, again, it is an infrared-unsafe change, since an arbitrarily soft emission has altered the hard jets. A collinear splitting produces the collinear-unsafe failure. Replacing one hard seed by two collinear particles, each carrying half its transverse momentum, may leave neither above the seed threshold, so that a jet which existed before the splitting is absent after it. Therefore, in both cases, the divergent emissions of equation (2.2) feed directly into the jets themselves, and the perturbative prediction diverges with them.

Going back to the family introduced before, the sequential algorithms avoid both failures by never relying on seeds or on fixed geometric cones. In these algorithms, jets are built by repeatedly merging inputs according to a distance measure, so that soft or collinear additions change only the order in which already nearby inputs are combined and never the final assigment of hard momentum jets. This structural difference is the reason why this entire family is IRC safe and it is why it is recommended for any measurement to be compared with theory over the cone algorithms.

A sequential recombination algorithm is defined by a distance measure between each pair of inputs and between each input and the beam. The generalised-$k_\mathsf{T}$ family [7] writes these as

$$d_{ij} = \min\left(p_{\mathsf{T},i}^{2p},\, p_{\mathsf{T},j}^{2p}\right) \frac{\Delta R_{ij}^2}{R^2}, \tag{2.6}$$

$$d_{iB} = p_{\mathsf{T},i}^{2p}, \tag{2.7}$$

where $\Delta R_{ij}^2 = (y_i - y_j)^2 + (\phi_i - \phi_j)^2$ is the squared angular separation of the inputs in rapidity $y$ and azimuth $\phi$, $R$ is the jet radius parameter, and the integer $p$ selects the algorithm. The algorithm, at each step, finds the smallest of all the distances. f it is a $d_{ij}$, the inputs $i$ and $j$ are recombined into a single input. If it is a $d_{iB}$, the input $i$ is declared a final jet and removed. This procedure repeats until no inputs remain. The returned jets are IRC safe for any integer value of $p$, but the value of $p$ changes extremely how the clustering proceeds. The three choices $p = -1$, $p = 0$ and $p = +1$ give, respectively, the anti-$k_\mathsf{T}$, Cambridge/Aachen and $k_\mathsf{T}$ algorithms, of which the first two are the tools this work relies on and are described in the following subsections.

In all three of those algorithms, the parameter $R$ appears and sets the angular reach of the clustering through the ratio $\Delta R_{ij}^2/R^2$ in equation (2.6). A smaller $R$ yields narrower jets that resolve nearby structure but also capture less of a single parton radiation. A larger $R$ collects more of that radiation at the price of admitting more contamination from unrelated soft activity. The most appropriate choice depends on what we intend to use the jets for, and the two algorithms this work relies on are used with different values of $R$ for correspondingly different goals. This point is, again, taken up in Sections 2.2.2 and 2.2.3.

### 2.2.2 The anti-$k_\mathsf{T}$ algorithm

The choice $p = -1$ in Eqs. (2.6) and (2.7) defines the anti-$k_\mathsf{T}$ algorithm [8]. With this value the inter-particle distance becomes

$$d_{ij} = \min\left(\frac{1}{p_{\mathsf{T},i}^2}, \frac{1}{p_{\mathsf{T},j}^2}\right)\frac{\Delta R_{ij}^2}{R^2}, \tag{2.8}$$

so that the distance scales with the inverse squared transverse momenta of the inputs. This makes the smallest distances, and thus the first recombinations, commit to the hardest particles. The clustering is therefore driven by the hard inputs. So, a hard particle accumulates the softer inputs around it before soft inputs cluster among themselves.

The name ”anti” of this algorithm reflects this inversion. In the $k_\mathsf{T}$ algorithm, the one with the choice of $p = 1$, the distance scales with the positive powers of transverse momentum, making the soft unputs cluster first and the jet being assembled from the bottom, following the inverse of the physical shower order. The anti-$k_\mathsf{T}$ algorithm does the reverse of it. When taking $p = -1$, the soft-first ordering of $k_\mathsf{T}$ turns into a hard-first one. Effectively, it is the $k_\mathsf{T}$ algorithm with the role of soft and hard interchanged, and the ”anti” part shows exactly that inversion. The two sit at opposite ends of the generalised family, with Cambridge/Aachen, free of any transverse momentum dependence, between them.

There is a geometric consequence because of this ordering. A hard particle with no other hard particle within an an angular distance $2R$ gathers all softer inputs within a distance $R$ of it, producing a jet that is exactly circular in the $(y, \phi)$ plane. When two hard particles are closer than $2R$, the border between their jets is set by their relative transverse momenta with the harder one retains the larger share, but the jets remain regular in shape. Due to soft radiation being last in the clustering, it cannot deform these borders. They are spread across the hard jets without chaging their shape.

The region of the $(y, \phi)$ plane from which a jet gathers its inputs is a useful way to describe this stability. This area can be probed operationally by adding a large number of infinitesimal soft particles to an event and record which jet each of these particles is assigned. For the majoroty of the algorithms, the resulting areas are irregular and fluctuate from jet to jet, because soft particles impact the clustering. However, for the anti-$k_\mathsf{T}$ algorithm, where soft particles are clustered last and cannot reshape the hard jets, the areas are close to the ideal value $\pi R^2$ and fluctuate little. A stable and expected area is what makes the subtraction of pile-up and underlying-event contamination tractable, since that contamination scales with the area over which each jet collects it.

This regularity is what made the anti-$k_\mathsf{T}$ algorithm the standard jet definition at the LHC. Since the jet border is stable against soft radiation, the area over which a jet collects contamination from pile-up and the underlying event is well defined and almost constant, and the corresponding energy corrections are simple to calibrate. Therefore, the anti-$k_\mathsf{T}$ algorithm combines the IRC safety guaranteed for the whole generalised-$k_\mathsf{T}$ family with a jet form that is simple to treat experimentally. This combination thus provides regular borders, which the other two generalised-$k_\mathsf{T}$ algorithms do not.

The importance of the anti-$k_\mathsf{T}$ is also a matter of history. When the LHC started taking data, an IRC safe jet definitions was needed, so that measurements could be compared with the increasingly precise perturbative calculations that became available. The algorithm was also required to be fast enough to run along the very large event samples which the LHC would produce. The anti-$k_\mathsf{T}$ algorithm, together with its efficient implementation in the FastJet package [9], met both needs, and it was adopted as the

baseline jet definition by the LHC experiments, a role it has held since.

It is because of this that the anti-$k_\mathrm{T}$ algorithm was used in this work to define the jets, with radius parameter $R = 0.8$. Jets constructed in this way are conventionally denoted as AK8. Howver, the algorithm does not provide a physically meaningful account of how the jet was assembled. This is becacause its sequence is ordered by hardness rather than by angle and the intermediate recombinations bear no correspondence to the sequence of physical splittings in a parton shower, which is ordered in angle. The clustering history of an anti-$k_\mathrm{T}$ jet is, in this sense, an artefact of the algorithm rather than a record of the underlying radiation. Recovering that history, which is what the substructure observables of this work require, calls for a different member of the family, applied to the constituents of the jet once it has been defined. That algorithm is the Cambridge/Aachen algorithm and it is the subject of the next subsection.

### 2.2.3 The Cambridge/Aachen algorithm

From equation 2.6 and equation 2.7, if one chooses $p = 0$, the Cambridge/Aachen (C/A) algorithm is defined [10]-[11]. With this choice, the transverse momentum factors drop out of the distance measure entirely, leaving the equation to a purely angular form:

$$d_{ij} = \frac{\Delta R_{ij}^2}{R^2}. \tag{2.9}$$

The clustering process with this algorithm proceeds by repeatedly merging the pair of inputs closest in angle, independently of how hard or soft they are. Where the anti-$k_\mathrm{T}$ algorithm is driven by the hardest inputs, the C/A algorithm orders them by angular separation alone. This means that the first merges are the most collinear and the last the widest.

This type of angluar ordering is what makes the C/A algorithm the optimal tool choice for evaluating jet substructure. As discussed in Section 2.1, the parton shower is ordered in angle, with the widest angle splittings happening earliest in the physical branching sequence. Therefore, the C/A algorithm mirrors, in reverse, the order in which the shower developed. By undoing the clustering process one step at a time, an approximation of the branching sequence is retraced from the widest splitting to the narrowest splitting, and each step exposes a single emission, characterised by the angular separation $\Delta R$ and the relative transverse momentum $k_t$ of the two branches it separates. This agreement between the C/A history and the physical shower is what the Lund jet plane, discussed in Section 2.3, is built to exploit, and it is the reason jet substructure observables are extracted from the C/A history rather than the anti-$k_\mathrm{T}$ one.

The declustering process can be asserted explicitly. Starting from the single C/A object that contains the whole jet, one undoes the last recombination, splitting it into two subjets $a$ and $b$, labelled so that $p_{t,a} > p_{t,b}$, meaning $a$ is the harder branch and $b$ the softer branch. This softer branch is taken as the emission and then, the declustering process, follows the harder branch $a$, repeating the previous step of splitting it. Iterating along the hard branch until it can no longer be declustered traces out the primary sequence of the jet. This is a ordered list of emissions, from the widest and earliest to the most collinear and latest.

Every splitting in this sequence is characterised by a small set of kinematic variables. The angular separation of the two branches $\Delta R_{ab}$, and their relative transverse momentum $k_t = p_{t,b}\,\Delta R_{ab}$, together locate the emission on the two scales that matter. This characterises the splitting by how hard and how hard it is. A third variable, the momentum fraction $z = p_{t,b}/(p_{t,a} + p_{t,b})$, computes how the momentum

was distributed between the emission and the emitter. These are the quantities from which the Lund jet plane of section 2.3 is constructed, and computing them at every declustering step is the immediate goal of the C/A history.

By following only the harder branch, at each step, one captures the primary emissions, although it discards the structure within the softer branches. In order to recover that structure, it is necessary to decluster it recursively. By choosing a primary emission, the softer branch $b$ is itself declustered in the same way, exposing the emission within it and generating a secondary sequence of emissions associated with that primary splitting. The primary sequence records how the leading parton radiated and the secondary sequences record how those radiated partons radiated in turn. In this work, both primary and secondary emissions' observables are needed in order to compare the orientation of a primary splitting with that of the secondary splitting it seeds.

In practice, the anti-$k_\mathrm{T}$ and C/A algorithms are applied in succession and complement each other. A jet is first defined with the anti-$k_\mathrm{T}$ algorithm, since it fixes its constituents and its regular boundary. Those constituents are then extracted and reclustered with the C/A algorithm, in order to obtain an angularly ordered history that can expose the emissions within a jet. In this work, the radius parameter of the C/A reclustering step was set to $R = 2$, larger than the anti-$k_\mathrm{T}$ value of $R = 0.8$, to ensure that all the constituents of the original jet are gathered into a single C/A object and that the declustering history is complete with no emission lost at the edge. Thankfully, the kinematics of the jet are unchanged by performing this operation because the same constituents with the same momentum are involved, however they now become organised into a sequence from which the physical splittings can be read.

The division of labour between the two algorithms is thus deliberate. The anti-$k_\mathrm{T}$ algorithm supplies a jet with a stable, regular boundary, well suited to the energy calibration and pile-up subtraction discussed in Section 2.2.2. The C/A algorithm applied to that jet's constituents supplies a history that mirrors the physical showering sequence with angularly ordered branching history which substructure requires. Neither algorithm alone would suffice. The anti-$k_\mathrm{T}$ algorithm gives a clean jet but an unphysical history, while the C/A algorithm gives a physical history but an irregular, contamination-prone boundary. Using each where it is strong is what makes the substructure analysis of this work both experimentally robust and physically interpretable.

### 2.2.4 Flavoured jets and the IFN algorithm

In the previous subsections, the algorithms cluster inputs on the basis of their kinematics alone, which is sufficient for most purposes. However, for the core measurement of this work, kinematics alone are not enough. As foreseen in Chapter 1, the goal is to explore the spin correlations of gluon splitting, and that requires distinguishing the channels of equations (2.4), (2.5) and (2.3), which produce, respectively, a pair of gluons, a quark or anti quark with a gluon and a quark-antiquark pair. The channels differ not only in their rates but also their spin interference signal carries different signs and magnitudes. Evaluating this effect at its real strengh thus require separating channels, and, in addition, that requires flavour assigment to the splittings inside a jet. This raises the obstacle of how can one assign flavour to a jet, or even to a branch inside one, in a way that remains theoretically well defined and experimentally robust.

The most obvious procedure is to sum the flavour of a jet's constituents, by counting a quark as $+1$ and an antiquark as $-1$ for each quark flavour separatelly, so that the net flavour of a jet is that of the parton which initiated it. Although this procedure is simple, it is not IRC safe and the root of the problem is precisely the $g \to q\bar{q}$ splitting. A gluon carries no net flavour, but it can split into a quark-antiquark pair that is soft or

wide in angle or even both. This quark and antiquark can then be assigned to different jets, or even to different branches of the same jet, making each acquire a net flavour that the gluon never possessed. Because the splitting can be arbitrarily soft, the missassigment occurs with a probability that diverges in the same manner as the emission spectrum of equation (2.2), and no virtual correction is assigned to cancel it. This is the same mechanism for kinematic observables, identified in Section 2.2.1 but now for the flavour. Here, this simple procedure is ruined by exactly soft and collinear radiation that QCD produces most abundantly, and a fixed order calculation of it fails to converge. Thus, flavour assigned sloppily is not a physical quantity and it cannot be compared with theory.

Restoring IRC safety to jet flavour has been pursued for two decades. The first safe prescriptions achieved IRC safety by making the clustering distance itself depend on the flavour of the objects being combined [12], which succeeded in supressing the divergence, but they costed the produced jets to not have the kinematics of the fundamental algorithm. This means that a flavour aware jet and its equivalent anti-$k_{\mathrm{T}}$ or C/A jet were no longer the same object. For a measurement that requires both kinematics of a jet and the flavour of its structure, that cost is too high. In this work, the approach adopted was using the Interleaved Flavour Neutralisation (IFN) [13], which was designed exactly to avoid that issue. IFN leaves the kinematics of the clusturing entirely untouched and treats flavour as a separate layer that runs alongside the normal clustering sequence. The jets, which are returned by this algorithm, are the identical to the C/A jets of Section 2.2.3. This includes constituents and momentums. What IFN gives in addition is a flavour label for each, computed by a procedure that is stable against soft and collinear emissions.

The principle behind this new procedure is neutralisation. While the clustering occurs, IFN tracks down the flavour carried by every object in the sequence, and, at each stage, asks wether a flavoured object can be matched to a nearby object of opposite flavour, which is the partner who would have originated the common $g \to q\bar{q}$ splitting. Whenever it finds such a pair, the two flavours are neutralised against each other, so that the ingenuine flavour is cancelled at the moment it would otherwise be passed up the clustering tree and inherited by a jet. This makes the quark and antiquark born of the same soft or collinear gluon splitting reunite in flavour, even though their kinematics remain whatever the C/A algorithm placed them. The matching is ruled by a flavour distance that weighs how soft and how collinear a candidate pair is, so that it is exactly the soft and collinear configurations that are neutralised, while genuinely distinct hard flavours are left in place.

What IFN brings to this work is that it does not merely give a prescription that labels the final jet. It exposes the flavour at every step of the clustering and not only at the end. This is the feature that the present analysis required. he clustering of a C/A history produces at each step a splitting of one object into two branches. IFN attaches to those branches a flavour label that is IRC safe and, therefore, meaningful at fixed order in perturbation theory. The abstract distinction between the discussed channels in Eqs. (2.4), (2.5) and (2.3) is thereby made concrete, and it can be read off at each splitting in the history of a jet. It is on this per splitting flavour that the channel selection at the heart of the measurement is built. The algorithm is distributed as a plugin to the FastJet package [9], and the configuration adopted here, together with the way its per-splitting labels are combined into the selection itself, is set out in Chapter 3.

## 2.3 The Lund jet plane

Described in Section 2.2.3, the declustering of the C/A history, links with every jet an ordered sequence of emissions, each carrying an angular separation $\Delta R$, a relative transverse momentum $k_t$ and a momentum fraction $z$. This sequence, on its own, is just a list of numbers. Jets have different numbers of emissions

and the raw variables reach many orders of magnitude. However, the Lund jet plane [1] can represent this list into something clear, when mapping each emission to a single point in a two-dimensional plane whose coordinates are chosen so that the structure of QCD radiation is exposed in its simple form.

### 2.3.1 Coordinates and the uniform density

The procedure to construct the Lund jet plane is direct. For each jet, the primary declustering of the C/A history, of Section 2.2.3, yields a list of emissions. Every emission in that list is mapped to a single point on the plane using its angular separation $\Delta R$ and relative transverse momentum $k_t$ as coordinates. When repeating this for every emission in a sample and overlaying the resulting points, it produces a two-dimensional distribution, which is then normalised to the number of jets inside the sample, giving the average density of emission per unit area. It is this averaged distribution, rather than the pattern of any single jet, that is referred to as the Lund jet plane and that carries the statistical information exploited below.

Each primary emission is placed on the plane at the coordinates:

$$\left(\ln \frac{1}{\Delta R},\, \ln k_{\mathsf{T}}\right), \tag{2.10}$$

the horizontal axis measures the inverse angle of the splitting and the vertical axis its hardness, both logarithmically. The choice of logarithmic axes is not cosmetic. Going back to Section **??**sec:qcd) that the emission probability of a gluon emission carries the double structure $d\mathcal{P} \propto \alpha_s\, d\theta/\theta\; dE/E$, which is one factor for the collinear enhancement and one for the soft enhancement. When written in terms of Lund coordinates of equation (2.10), each of these factors becomes flat. The measure $d\theta/\theta$ is simply $d\ln(1/\Delta R)$, and the energy factor maps onto $d\ln k_{\mathsf{T}}$. The doubly divergent emission spectrum of the raw variables therefore becomes, on the Lund plane, an approximately uniform density of emissions per unit area,

$$\rho \;=\; \frac{1}{N_{\text{jet}}}\,\frac{dn_{\text{emissions}}}{d\ln(1/\Delta R)\; d\ln k_{\mathsf{T}}} \;\approx\; \text{const} \times \frac{\alpha_s}{\pi}, \tag{2.11}$$

up to the slow running of $\alpha_s$ and the effects discussed below. It is this flatness that makes the plane so useful against a uniform background. Any physical effect beyond the leading soft-collinear approximation therefore appears as a localised enhancement or suppression, rather than being obscured by the rapidly falling raw spectrum. This uniform density is precisely what is observed in practice. Figure 2.1 shows the average primary Lund plane obtained from a simulated sample of QCD jets. Through the perturbative majority of the plane the density is approximately constant, as anticipated by equation (2.11), falling away only towards the boundaries of the kinematically accessible region.

### 2.3.2 Regions of the plane

The reachable area of the plane is delimited, and its interior can be divided into different regions with distinct physical source, sketched in Figure 2.2. The upper boundary is set by the kinematics of the jet. Every emission cannot be harder than the jet which contains it, so it caps $\ln k_{\mathsf{T}}$ from above. Along the right of the plane, at large $\ln(1/\Delta R)$, lie the collinear emissions, those radiated at small angle to their

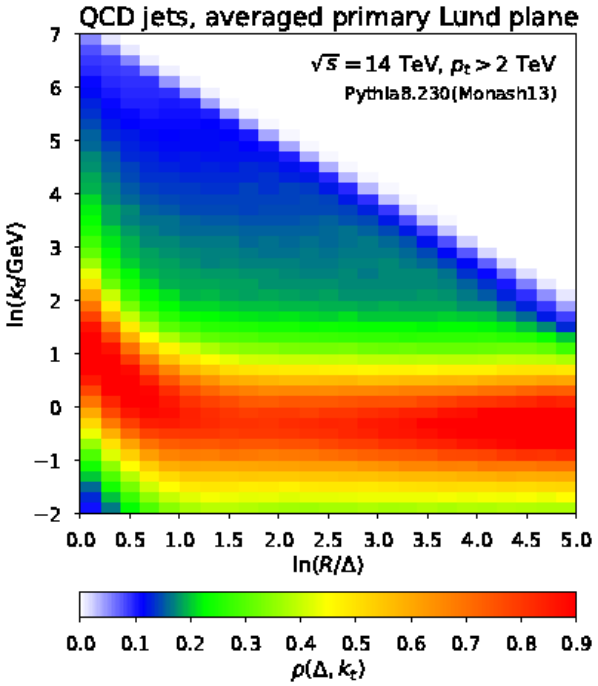



**Figure 2.1** – The average primary Lund plane density, ρ, for jets clustered with the C/A algorithm and $R = 1$ having $p_t > 2$ TeV and $|y| < 2.5$, in a simulated QCD dijet sample [1].

emitter. Towards the bottom, at small ln $k_\mathrm{T}$, lie the soft emissions. The perturbative description of Section 2.1 holds across the majority of this region, where $k_\mathrm{T}$ is large enough for $\alpha_s$ to remain small, making the density there be the near-constant value of equation (2.11).

The exceptional region in the lower-right corner, at both small angle and low transverse momentum. In that region, $k_\mathrm{T}$ falls to the scale of a few hundred MeV, $\alpha_s$ grows large, and the perturbative description breaks down. This region is known as the non-perturbative region and it is populated by hadronisation and by the soft, wide-angle contamination present inside the event. This way, one strong feature of the Lund jet plane is that these non-perturbative effects are confined to a restricted corner of the plane, clearly separated from the perturbative part and not mixed with the whole observable. A measurement can therefofore restrict, by eye or by an specific cut, to the region where theory is applicable.

### 2.3.3 Grooming and the primary and secondary planes

This clear separation motivates grooming methods. Grooming is the intentional removal of soft, wide-angle emissions that dominate non-perturbative regions of the Lund jet plane. The grooming method used in this work is Soft Drop [14], which walks through the declustering and, at each step, keeps the harder branch unless the softer one satisfies the following:

$$z > z_\mathrm{cut} \left( \frac{\Delta R}{R} \right)^{\beta}, \tag{2.12}$$

in which case both branches are kept and the declustering precedure continues. The parameter $z_\mathrm{cut}$ specifies how soft an emission must be to be discarded, and $\beta$ manages the angular reach of the grooming. The choice $\beta = 0$, adopted here for the Soft Drop in this work, makes the condition purely one of momentum fraction , $z > z_\mathrm{cut}$, independent of angle. In this condition, Soft Drop becomes a modified

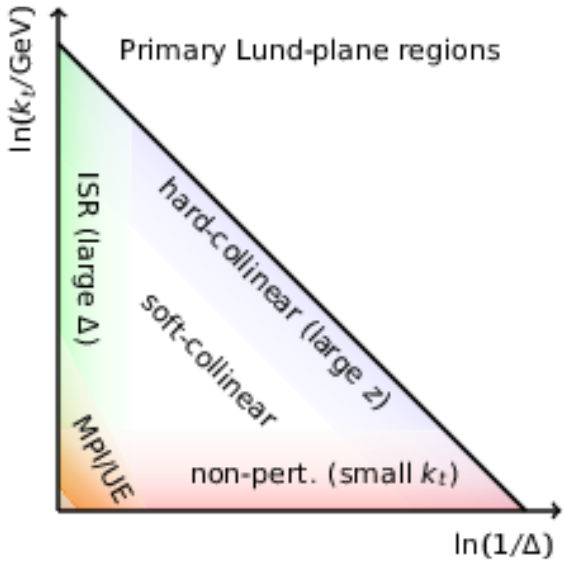



**Figure 2.2** – Schematic representation of the di□erent regions of the Lund plane [1].

Mass-Drop Tagger [15]. Applying this grooming method becomes very visible on the Lund jet plane because it removes the emissions in the soft, low $k_{\mathrm{T}}$ band, clearing the non-perturbative corner and leaving a plane populated only by the harder splittings.

In the present work, two versions of the Lund jet plane are built and used as follows. The primary Lund plane is built with the emissions along the hard branch of the jet and describes how the leading parton radiated. The secondary Lund plane is built by taking the chosen primary emission and declustering its softer branch in turn, using the same method as the primary Lund plane, as described in Section 2.2.3. The resulting points describe how that radiated parton itself went on to radiate. The two planes together deliver complementary information and both are needed for the spin observables of Section 2.4, which compare the orientation of a primary splitting with that of the secondary splitting it seeds. Figure 2.3 illustrates these two levels of structure for two example jets. The middle row shows the full Lund diagram, in which every emission, primary and secondary, is represented, while the bottom row shows the primary Lund plane alone, retaining a single point for each primary splitting.

## 2.4 Collinear spin correlations

In the previous sections, the tools assembled describe a parton shower as a sequence of independent splittings, each blind to those before it. This description is incomplete since intermediate parton in a shower carry spin and their spin is correlated with the kinematics of the splitting that produces them. Furthermore, when that parton branches, that correlation is transmitted to the azimuthal orientation of its own decay products. So, successive splittings are therefore not independent, meaning that the plane of one branching is correlated with the plane of the next. These are the collinear spin correlations which this work presents, and they are the subject of this section.

This correlation effect is a quantum interference phenomenon, with no counterpart in a classical picture of sequential radiation, and it shows in a distinctive and robust way. This phenomenon manifests as a modulation of the azimuthal angle between successive splitting planes with a $\pi$ periodicity instead of $2\pi$, varying directly as a function of $\cos 2\psi$. This section develops that statement in four steps. Section 2.4.1

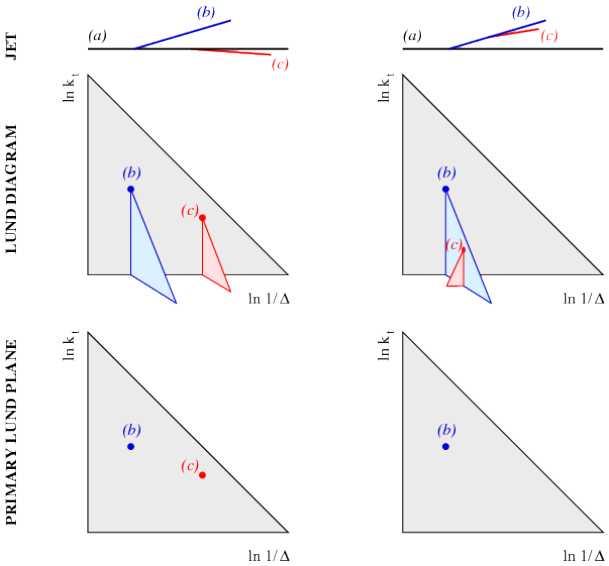



**Figure 2.3** – Different representations for two jets. Top: the particles inside the jet. Middle: the full Lund diagram. Bottom: the primary Lund plane [1].

shows the physical mechanism and the origin of the $\cos 2\psi$ modulation. Section 2.4.2 shows how the size and sign of the modulation depend on the splitting channel, the property that makes flavour identification essential to the measurement. Sections 2.4.3 and 2.4.4 then introduce the two observables through which this work accesses the effect. These observables being the Lund declustering angle $\Delta\psi_{12}$, computed from the branching history of Section 2.3, and the three-point energy correlator, which reaches the same physics through the flow of energy within the jet.

### 2.4.1 Mechanism and the $\cos 2\psi$ modulation

The origin of the correlation lies in the spin of the intermediate gluon [16]. Consider a gluon produced in a first splitting that subsequently undergoes a second splitting of its own. This corresponds to the setup of the primary and secondary Lund planes discussed in Section 2.3. Since the gluon is a spin-one particle, a hard, nearly on-shell gluon possesses only two physical polarisation states, both transverse to its direction of motion. In the collinear limit, these states are most naturally described not as circular polarisations, but as two orthogonal linear polarisations, each representing an oscillation of the field along a fixed axis in the plane transverse to the gluon momentum.

The polarisation of the intermediate gluon is not random. The first splitting, which produced it, selects a preferred transverse axis. Therefore, the gluon emerges linearly polarized with its polarisation correlated with the plane of that first branching. This is the main and essential point. A gluon radiated in a splitting carries a memory of the orientation of that splitting, engraved in the direction of its linear polarisation.

Afterward, that memory is read out by the second splitting. The probability that a linearly polarised glluon branches into a given final state depends on the azimuthal orientation of the second splitting plane in respect with the polarisation axis. Writing $\psi$ for the angle between the first and the second splitting planes, the differential rate acquires a dependence on $\psi$ through this alignment, and the two splittings become correlated in their relative azimuth.

The period of that correlation is fixed by the nature of the polarisation. A linear polarisation is specified by an axis, not by a direction along it. Rotating the polarisation by $\pi$ returns the same physical state,

since an oscillation along an axis is unchanged by reversing its sense. The azimuthal dependence must therefore be invariant under $\psi \to \psi + \pi$, which forbids a $\cos\psi$ term and admits $\cos 2\psi$ as its leading harmonic. This is the distinctive signature of the effect. The result has to have a modulation of period $\pi$ rather than $2\pi$, reflecting the headless, axial character of linear polarisation. Collinear spin correlations announce themselves not as any azimuthal dependence, but specifically as a $\cos 2\psi$ dependence.

At second order in the strong coupling and in the collinear limit, the cross section differential in $\psi$ therefore takes the form

$$\frac{d\sigma}{d\psi} \propto a_0 \left(1 + \frac{a_2}{a_0} \cos 2\psi\right), \tag{2.13}$$

where $a_0$ is the spin-averaged rate, or, in other words, the result one would obtain by ignoring the polarisation of the intermediate gluon, and $a_2$ is the coefficient of the interference term [16]. The ratio $a_2/a_0$ measures the strength of the correlation where it vanishes if the intermediate gluon carries no polarisation memory, and its magnitude sets the depth of the $\cos 2\psi$ modulation. Its sign determines the phase, wether the second splitting preferentially aligns with the first splitting plane, giving maxima at $\psi = 0, \pi$, or preferentially lies perpendicular to it, giving maxima at $\psi = \pm\pi/2$. The measurement of these spin correlations reduces, in essence, to the measurement of this single ratio, and the whole of the analysis that follows is directed towards extracting it cleanly.

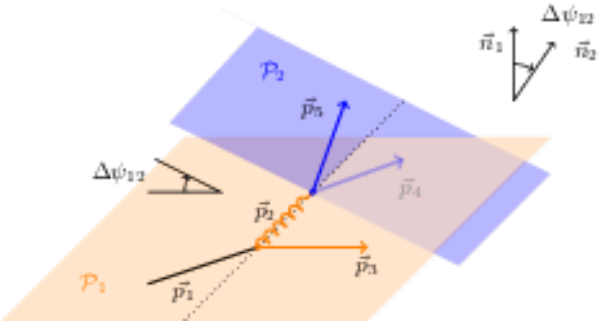



**Figure 2.4** – Azimuthal angles are defined between successive splitting planes for the $1 \to 23$ splitting, $\mathcal{P}_1 \supset \{\vec{p}_2, \vec{p}_3\}$, and the $2 \to 45$ splitting, $\mathcal{P}_2 \supset \{\vec{p}_4, \vec{p}_5\}$. The figure also depicts the vectors normal to the two planes, $\vec{n}_1$ and $\vec{n}_2$ [16].

### 2.4.2 Channel dependence of $a_2/a_0$

The process described in Section 2.4.1 was made for a generic intermediate gluon, however, the strength and sign of the resulting modulation are not universal. These parameters depend on the nature of the second splitting, or, in other words, on the channel through which the polarised gluon decays. The two channels available are those of equations (2.4) and (2.5). The gluon can spit into a pair of gluons, $g \to gg$, or into a quark–antiquark pair, $g \to q\bar{q}$. Each of these channels read out the polarisation of the parent gluon, but each one of them do so with opposite sign and different strengh [16].

The reason behind this difference comes from the spin of the produced particles. The angular The angular distribution of a splitting depends on how the daughters' spins couple to the polarisation axis of the parent, and this coupling differs between a final state of two vector gluons and one of two spin-half fermions [16]. The difference is that the interference coefficient $a_2/a_0$ of equation (2.13) appears with one sign for the $g \to gg$ channel and the opposite sign for the $g \to q\bar{q}$ channel, where one channel prefers the alignment of the two splitting planes, the other a perpendicular configuration.

This sign difference is the main goal which this work is built to overcome. If one does a measurement without distinguishing the two channels of the second splitting, the contributions of both sum together, leaving a result which almost cancels itself. The endured signal measured in this case is not the intrinsic strength of the correlation but a residue, because its magnitude and even its sign are set by the balance between the $g \to gg$ and $g \to q\bar{q}$ channels present in the sample. However, this interference is not absent. It is just diluted and this dilution reflects the flavour composition of the selection rather than any property of QCD.

Thus, it is required to isolate a single channel in order to make a reliable measurement of the effect. This work shows the selection procedure of the $g \to q\bar{q}$ splittings, and measures the parameter $a_2/a_0$ on that subsample alone, replacing the diluted inclusive residue with the coefficient of a single channel. This is precisely what the infrared-safe and collinear-safe flavour labelling of Section 2.2.4 makes possible. IFN tags the flavour, exposed during the declustering, of each splitting, allowing the $g \to q\bar{q}$ channel to be enriched and the $\cos 2\psi$ modulation to be measured as a function of the purity of that enrichment. The channel dependence described here is therefore not just a feature of the physics but the reason the flavour selection developed in this work is required, and it is the procedure that connects the spin observable of the next section to the flavoured-jet algorithm of Section 2.2.4.

### 2.4.3 The Lund observable $\Delta\psi_{12}$

The effect described in the previous sections can be studied using a concrete observable, given by the Lund declustering of Section 2.3, in order to indentify the two splittings whose planes are to be compared. This construction was proposed in [16], and the observable is the azimuthal angle $\Delta\psi_{12}$ between the hardest primary and secondary branchings of a jet.

The two splittings are selected as it follows. When declustering the primary Lund sequence of a jet, one selects the splitting which passes the Soft Drop condition of equation (2.12) and has the largest relative transverse momentum $k_\mathrm{T}$ among all the primary sequence. This splitting is then called the primary splitting. Then, declustering the soft branch of this selected primary splitting, we build the secondary sequence generated by it. Thus, with the same procedure as in the primary sequence, one selects the splitting which passes the Soft Drop condition of equation (2.12) and has the largest relative transverse momentum $k_\mathrm{T}$ among all the secondary sequence. This splitting is called the secondary splitting. Each of the two chosen splittings defines a plane in momentum space. The plane $\mathcal{P}_1$ spanned by the branches of the first, and $\mathcal{P}_2$ spanned by the branches of the second. With this, $\Delta\psi_{12}$ is the azimuthal angle between them, illustrated in Figure 2.4.

By construction, $\Delta\psi_{12}$ is the realisation for jets of the abstract angle $\psi$ of equation (2.13). Its distribution therefore inherits the $\cos 2\psi$ modulation derived in Section 2.4.1,

$$\frac{1}{\sigma}\frac{d\sigma}{d\Delta\psi_{12}} \;\propto\; 1 + \frac{a_2}{a_0}\cos 2\Delta\psi_{12}, \tag{2.14}$$

and the coefficient $a_2/a_0$ extracted from it is the quantity that carries the spin-correlation physics. Two features of the definition are essential to what follows. First, because the second splitting is drawn from the secondary sequence seeded by the first, $\Delta\psi_{12}$ compares precisely the two successive branchings between which the polarisation memory of Section 2.4.1 is transmitted. Second, when that second splitting is the decay of an intermediate gluon, its channel $g \to gg$ or $g \to q\bar{q}$ can be identified by the flavour labelling of Section 2.2.4. Splittings of the form $q \to qg$, in which no intermediate gluon is present,

carry no correlation and act only as a background to be removed. It is this flavour resolution that makes $\Delta\psi_{12}$ not merely a measurement of the inclusive, diluted modulation, but a channel-resolved probe of the effect. By restricting to secondary splittings tagged as $g \to q\bar{q}$, and thereby separating them both from $g \to gg$ splittings and from the correlation-free $q \to qg$ background, the sign cancellation of Section 2.4.2 is undone, and $a_2/a_0$ can be measured on a single channel.

### 2.4.4 The three-point energy correlator

Although the lund observable described in the previous section gives a result for the collinear spin correlations of Section 2.4.1, it is not the route to do so. The $\cos 2\psi$ modulation can be accessed through the energy flow within a jet and without any reference to any clustering history, using the three-point energy correlator (EEEC) [17]. Where $\Delta\psi_{12}$ reconstructs the two splitting planes by declustering, the energy correlator reads the effect directly from the angular and energy distribution of the jet's constituents, and the two therefore provide complementary handles on the same underlying physics.

The EEEC measures the energy-weighted distribution of triplets of particles as a function of their relative angles. Concretely, it is defined as the energy-weighted sum over all triplets of particles in the jet,

$$\frac{1}{\sigma}\frac{d^3\Sigma}{d\psi\, d\theta_S\, d\theta_L} = \sum_{i,j,k} \frac{p_{\mathrm{T},i}\, p_{\mathrm{T},j}\, p_{\mathrm{T},k}}{p_{\mathrm{T,jet}}^3}\, \delta(\psi - \psi_{ijk})\, \delta(\theta_S - \theta_{S,ijk})\, \delta(\theta_L - \theta_{L,ijk}), \tag{2.15}$$

where the sum runs over all triplets $(i, j, k)$ of the jet's constituents, each weighted by the product of their transverse momenta normalised to that of the jet. Here $\theta_S$ and $\theta_L$ are the smallest and largest opening angles of the triplet, and $\psi$ is the azimuthal angle between the plane spanned by the two closest particles and that spanned by the third. Its power for the present purpose is that this last variable retains sensitivity to the azimuthal orientation discussed in Section 2.4.1. This means that the correlator's dependence on $\psi$ carries the same $\cos 2\psi$ modulation, governed by the same interference between gluon polarisation states. Being built from energy weights and angles alone, the correlator is infrared and collinear safe by construction and requires no declustering, no grooming and no jet-flavour assigment.

This makes the energy correlator an independent cross-check on the physics extracted through $\Delta\psi_{12}$. It reaches the same $\cos 2\psi$ effect by a route that shares none of the reconstruction choices on which the Lund observable depends, so that agreement between the two would establish that the measured modulation is a property of the radiation rather than an artefact of any one analysis chain. Note that the coefficient $a_2/a_0$ extracted from the energy correlator is not expected to coincide numerically with that from $\Delta\psi_{12}$, since the two observables define the azimuthal angle through different objects and carry different sensitivities to the kinematic selection. What the two share is the sign and the qualitative behaviour dictated by the underlying gluon-polarisation interference, and it is at that level that they cross-check one another. The correlator is developed in this work as a complementary probe alongside the primary $\Delta\psi_{12}$ measurement, and its implementation and results are presented in Chapter 4.

# 3 DATA AND ANALYSIS

The preceding chapter established the observables through which collinear spin correlations can be accessed. Following that, this chapter describes how they are measured in practice and presents the data on which this analysis is built, the software environment in which it runs, and the sequence of steps that leads from the raw available events to the azimuthal distributions of Chapter 4. The goal of this chapter is to reveal all the steps necessary to make this analysis, securing reproducibility. That said, enough detail is given, and the code used is available publicly in [18], so that the analysis can be repeated in full from public inputs alone.

The chapter is organised as follows. Section 3.1 introduces the CERN Open Data initiative and the principles that make an analysis of this kind possible outside the experimental collaborations. Section 3.2 describes the relevant features of the CMS detector, and Section 3.3 the data format and the specific dataset used. Section 3.4 presents the analysis pipeline, from the extraction of jet constituents to the computation of the spin observables. Section 3.5 describes the flavour selection that isolates the $g \to q\bar{q}$ channel, and Section **??** assesses the systematic uncertainties affecting the measurement.

## 3.1 CERN Open Data

The data analysed in this work were obtained through the CERN Open Data Portal [2], the platform through which the LHC experiments release reconstructed collision data and the associated simulation to the public. Launched in 2014, the portal was the first initiative of its kind in high-energy physics, and it has since grown into a major repository holding data from the ALICE, ATLAS, CMS and LHCb collaborations. The first releases came from the CMS experiment, and it is CMS data that this work uses.

The CERN Open Data Portal does not provide detector output but data at the same level of reconstruction used in collaborations' own published analysis. These datasets include physics objects such as jets, tracks and reconstructed particles, with the calibration already applied and run quality information required to use them correctly. Alongside the collision data, the portal also releases the corresponding Monte Carlo simulations, the software environment necessary to analyse the datasets and documentation describing their format and content. All this features of the portal make it possible for any user to use their data for their own analysis and interpret the results.

The importance of this portal in this work is twofold. First, as made clear in this chapter's introduction, it makes the measurement reproducible. Every input from the collision and respective simulated datasets are publicly available, so that the analysis can be repeated in full by anyone. Second, the portal specifies, for each dataset, the steps required to use them correctly. For each dataset used in this work, CMS provides a list of available and validated runs, with the proper calibrations and standard energy corrections and residual effects, applied by them.

This type of analysis are feasible outside the collaborations has been demonstrated by a growing body of work built on CMS Open Data. Studies using released CMS samples have addressed problems ranging from the discrimination of quark and gluon initiated jets [19] to the measurement of $N$-point energy correlators inside jets [20], the latter closely related to the observables of the present work. These analyses establish both that public data retain the information needed for substructure measurements

and that results of genuine physics interest can be extracted from them. The present work is situated in this line. It applies the same public data to the measurement of collinear spin correlations, an effect not previously accessed in this way.

The only small downside of working with public data is the choice of software environment. The released CMS data are stored in formats tied to the experiment's reconstruction software, and reading them requires a compatible environment. CMS provides this in the form of a dedicated container image, inside docker [21], that reproduces the software stack in which the data were generated. In this work that environment was used to read the released files and extract the jet and constituent information required by the analysis, which was then carried out in a lighter, independent environment. The details of this arrangement are given in Section 3.3 and Section 3.4.

## 3.2 The CMS detector

The collision data used and analysed in this work was recorded by the Compact Muon Solenoid (CMS). This giant detector, alongside with ATLAS, is a general-purpose detector, at the LHC [22]. It is a cylindrical apparatus with 21.6 m in length and 14.6 m of diameter, built in concentric layers around the beam pipe with the characteristic feature of having a superconducting solenoid, with 6 m in internal diameter, capable of producing a uniform 3.8 T magnetic field, that bends the trajectories of charged particles allowing for their momenta to be measured. Its layers are capable of capturing relevant features of this analysis and they will be described below in the order of a particle transversing them.

The CMS coordinate system has the origin centered at the nominal collision point inside the detector, its y-axis points vertically upwards and the x-axis points radially inwards towards the center of the LHC. This makes the z-axis point along the beam axis towards Jura mountains from LHC Point 5 [22]. The polar angle $\theta$ is measured from the $z$ axis, The azimuthal angle $\phi$ is measured from the x-axis in the x-y plane and the radial coordinate in this plane is denoted by $r$. The polar angle $\theta$ is measured from the $z$ axis, although it is conventional to use the pseudorapidity $\eta = -\ln\tan(\theta/2)$, because differences in $\eta$ are invariant under boosts along the beam. The transverse momentum is denoted as $p_\mathrm{T}$ and it is the component of momentum in the plane perpendicular to the beam. These are the variables which jets and their constituents are used on this work.

Starting from the layer closest to the interaction point, the inner-tracker is an all-silicon system of pixel and strip detectos covering $|\eta| < 2.5$. This layer is placed inside the solenoid's field and it reconstructs the trajectories of charged particles, measures their momenta and detects the primary interaction vertex. Since substructure observables used in this work are built from the trajectories and momenta of a jet's charged constituents, this tracker is the subdetector which these measurements directly rely on.

There are two calorimeters surrounding the inner-tracker. The electromagnetic calorimeter (ECAL) uses lead tungstate scintillating crystal to measure the energy of electrons and photons. Surrounding it, the hardron calorimeter (HCAL), a brass/scintillator device, measures the energy of the hadrons. Together, these two calorimeters cover the central region and, with more calorimeters forward extending the coverage to $|\eta| < 5$, give CMS a near complete solid-angle coverage needed to measure the flow of energy in an event. The muon system outside the solenoid, plays no important role for this analysis.

When CMS reconstructs jets, they don't rely on just one of these subdetectors isolated. Instead, they use their own particle-flow algorithm, tuned specifically for the CMS detector [23], which combines signals from all the subdetectors, including tracks from the inner-tracker and energy deposits from the

calorimeters, to to reconstruct a list of individual particles in each event, the particle-flow candidates. It is these candidates that are the inputs to the jets' clustering algorithms of Section 2.2.1, and their charged component measured by the tracker that most directly determines the directions from which the substructure observables are built. This particle-flow algorithm is therefore the point at which the detector described here connects to the analysis that follows, and the data format in which its output is released is the subject of the next section. The following figure 3.1, gives a solid perspective of the CMS detector.

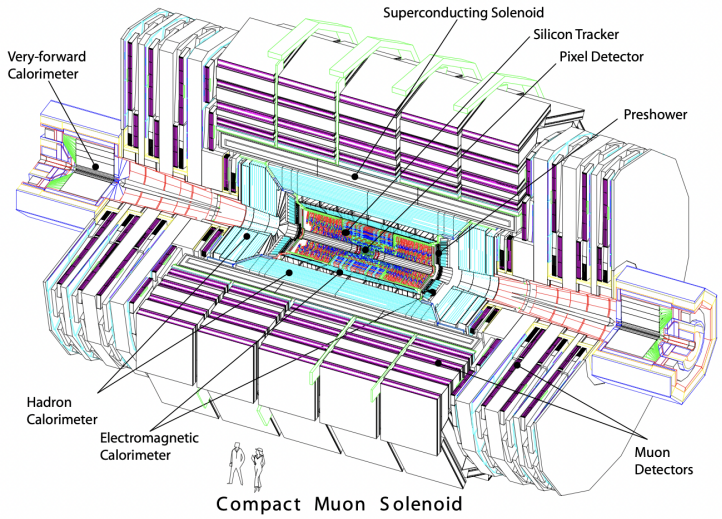



**Figure 3.1** – A perspective view of the CMS detector [22].

## 3.3 Data format and datasets

The CMS released reconstructed data in the CERN Open data format comes in a hierarchy of formats. Each format sacrifices completeness for size making the choice of format the first decision required for an analysis using their datasets, since it fixes what information is available and what variables are at hand. The three formats relevant here are summarised in Table 3.1.

**Table 3.1** – The three CMS data-tier formats, in decreasing order of size and detail. Sizes are approximate and quoted relative to the baseline AOD.

| Format | Content | Size |
|---|---|---|
| AOD | High-level physics objects plus substantial lower-level reconstruction detail. | baseline |
| MiniAOD | Analysis-level physics objects similar to AOD but slimmed: many collections are reduced, extra information is packed into lighter objects, and less commonly used low-level detail is removed. | $\sim 1/10$ of AOD |
| NanoAOD | Compact ntuple-style content: a limited set of per-object observables and precomputed IDs; little to no low-level detector detail (e.g., Particle Flow candidates dropped). | $\sim 1/20$ of MiniAOD |

For this work, the requirement is having access to the constituents of a jet. The substructure observables of Chapter 2 are built by declustering a jet into its constituents, so this analysis requires the individual particle-flow candidates that build the jet and not just a summary of the jet's kinematics. This immediality invalidates the NanoAOD data format since the particle-flow candidates are not stored there. The AOD format stores the particle-flow candidates, however it also stores other low-level reconstruction details

which this analysis does not need making it occupy a large volume of space. This leaves the MiniAOD data format, which occupies the middle ground of space and still stores particle-flow candidates required for substructure while discarding the low-level information that is not needed. Therefore, it is for this reason that the MiniAOD data format was the choice made for the format to be used throughout this work.

Between all the MiniAOD datasets, this work analysed the JetHT primary dataset from the 2016 data-taking period, in MiniAOD format [24]. Its characteristics are collected in Table 3.2. The data are proton–proton collisions at a centre-of-mass energy of $\sqrt{s} = 13$ TeV, recorded during run period G of 2016 and released as part of the Ultra-Legacy reprocessing, the final and most complete calibration of the Run 2 data.

**Table 3.2** – Characteristics of the JetHT primary dataset [24].

| Property | Value |
|---|---|
| Dataset | `/JetHT/Run2016G-UL2016_MiniAODv2-v2/MINIAOD` |
| Collision | proton–proton, $\sqrt{s} = 13$ TeV |
| Run period | 2016G (runs 278820–280385) |
| Events | $1.21 \times 10^8$ |
| Size | 4.1 TiB (1236 files) |

The JetHT dataset is well suited for this work analysis since it is enriched in jet events. CMS records events online that pass a two-level trigger system, composed by a hardware Level-1 trigger followed by a software High Level Trigger (HLT) [25]. The events of this dataset are then those accepted by a set of HLT paths, which require either a large scalar sum of jet transverse momenta or the presence of one or two energetic jets. As a result the sample is dominated by the high-transverse-momentum, high-multiplicity final states in which jet substructure is most informative. Since the dataset is served by triggers based on large-radius jets, events are required to pass one of the `HLT_AK8PFJet` paths. The analysis is restricted to jets with transverse momentum above $700$ GeV, where the unprescaled paths of highest threshold, `HLT_AK8PFJet450` and `HLT_AK8PFJet500`, are almost fully efficient. In this region any jet passing the selection also satisfies these paths, so the trigger requirement introduces no turn-on bias, and the sample sits in the hard, perturbative regime motivated in Section 2.1.

Furthermore, as required for all analysis of this dataset, the data is restricted to a list of runs certified as good physics, provided with the dataset release. So, events not in these certified runs do not enter this analysis. In addition, jets in this released sample do not require any energy calibrations to be applied to them, since they already carry the standard jet energy corrections applied by CMS in the Ultra-Legacy reprocessing.

Alongside the JetHT collision dataset, this analysis uses a simulated sample of QCD multijet events. This is because the flavour of a splitting inside a jet cannot be accessed in data, so it is not possible to determine the secondary splitting channel. What the simulation provides is the set of generated partons, carrying flavour, which are clustered in the same way as the reconstructed particles in data. First the partons are clustered into jets with the anti-$k_\mathrm{T}$ algorithm of section 2.2.2 and then, through the Cambridge/Aachen algorithm of Section 2.2.3 and the IFN algorithm of Section 2.2.4, into a branching history in which each splitting acquires a flavour. It is from this process that the channel of each splitting, $g \to q\bar{q}$, $g \to gg$ or $q \to qg$, is labelled. This flavour labelling process and how it is further used in collision data is discussed with more detail in Section 3.5. The simulated samples are QCD multijet datasets generated with Pythia 8 [26] using the CP5 tune [27], passed through the full simulation of the CMS detector and released in the same format as the collision data (MiniAODSIM). They are produced in intervals of the partonic transverse momentum $\hat{p}_\mathrm{T}$. The interval $800 < \hat{p}_\mathrm{T} < 1000$ GeV [28] is used to train the flavour selection, and the independent interval $600 < \hat{p}_\mathrm{T} < 800$ GeV [29] to compute its purity.

From these two datasets, there are some quantities extracted from each event, which are listed in Table 3.3. From each jet in collision datasets, their constituents are stored, since they are needed for the declustering of Chapter 2 and to build the Lund jet plane. The electric charge of each constituent is extracted in addition to its kinematics, as it is used later in the flavour selection of Section 3.5. In the simulated datasets, the same jet and constituent quantities are extracted, together with the final-state generated partons of the event, each with its identity. These partons carry the flavour information from which the splitting flavour is determined, as described in Section 3.5.

**Table 3.3** – Quantities extracted from each event. The jet and constituent quantities are stored for both the collision data and the simulation. The generated partons are available in the simulation only.

| Level | Quantity | Description |
|---|---|---|
| Jet | `jet_pt`, `jet_eta`, `jet_phi`, `jet_mass` | Transverse momentum, pseudorapidity, azimuth and mass of each jet |
| Constituent | `const_pt`, `const_eta`, `const_phi`, `const_mass` | Kinematics of each jet constituent |
| | `const_charge` | Electric charge of each constituent |
| Parton | `gen_pt`, `gen_eta`, `gen_phi`, `gen_mass` | Kinematics of each final-state parton (simulation only) |
| | `gen_pdgId` | Identity of each parton (simulation only) |

The jets themselves are taken from the `slimmedJetsAK8` collection, which are the anti-$k_\mathrm{T}$jets with $R = 0.8$ stored in MiniAOD, in a way that the jet definition of Section 2.2.2 is applied by CMS, making it not necessary to repeat this step in this analysis. Beyond the requirement of the transverse momentum described earlier in this section, jets are required to lie within $|\eta| < 1.7$, keeping them in the central, well-instrumented region of the detector. Events are further required to contain at least two such jets separated in azimuth by more than $2$ radians, selecting the back to back dijet topology in which the two hard jets recoil against one another. In addition jet's constituents are required to have $p_\mathrm{T} > 1$GeV in order to be recorded along with the jet. This selection follows closely that of the CMS measurement on which the present analysis is based [30], applied here to the earlier public dataset. The complete set of requirements is summarised in Table 3.4.

**Table 3.4** – Event and jet selection. The trigger and certified-run requirements apply to the collision data only.

| Requirement | Value |
|---|---|
| Trigger | `HLT_AK8PFJet` (data only) |
| Certified runs | validated-run list (data only) |
| Jet collection | `slimmedJetsAK8`, anti-$k_\mathrm{T}$ $R = 0.8$ |
| Jet $p_\mathrm{T}$ | $> 700$ GeV |
| Jet $\|\eta\|$ | $< 1.7$ |
| Dijet topology | at least two jets with $\Delta\phi > 2$ rad |
| Constituent $p_\mathrm{T}$ | $> 1$ GeV |

## 3.4 Analysis pipeline

The analysis made in this work, proceeds in a sequence of steps that start from the collision and simulated datasets, described in Section 3.3, and end with the final azimuthal distributions. These steps are split into two parts, separated by the software environment in which each one runs. The first part is the extraction of the jet's properties and constituent information from the datases. This step, as idicated by CMS, must be done inside the CMS Open Data container or the CMS Virtual Machine , since reading the MiniAOD

format requires the experiment's own reconstruction software. In this work, this step was done using the CMS Open Data docker container (CMSSW) with, for each dataset, the recommended version of it. The second step covers declustering of the jets and the computation of the observables. It is done in a separate environment built on ROOT [31], FastJet [9] and Python. This separation makes the user able to use the CMSSW software just for a single step which is completely mandatory and allows the rest of the analysis to run in a modern and portable environment.

The stages of the pipeline are ilustrated in Figure 3.2 and described in the following subsections. Section 3.4.1 describes the extraction of the jet ntuples from the collision and simulated samples. Section 3.4.2 describes the reclustering and declustering of each jet and the construction of the Lund observables and of $\Delta\psi_{12}$. Section 3.4.3 describes how the same procedure is applied to the generated partons in simulation in order to label the channel of each splitting. Section 3.4.4 describes the computation of the three-point energy correlator. The outputs of these stages feed the flavour selection of Section 3.5 and the measurement presented in Chapter 4. Since the datasets are extremely large, with a large number of jets and its variables, every stage has parallelisation integration. For the most demanding step of this analysis, which was the extraction of the variables, the code was run with the LIP's computing cluster, making it an hige difference for the time required to complete this step. The code used is publicly available [18], so that each stage can be reproduced from the released data alone.

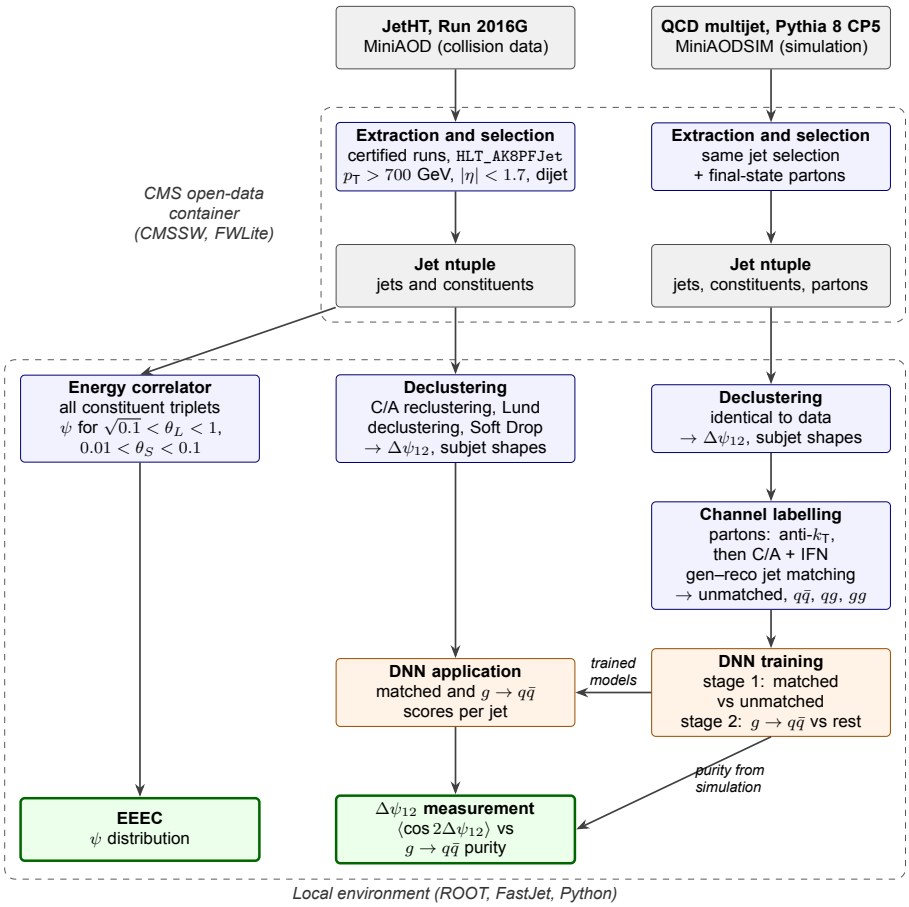



**Figure 3.2** – Schematic of the analysis pipeline. The extraction stage runs inside the CMS environment and produces the jet ntuples. The declustering, the generator-level labelling and the energy-correlator computation run outside it, and feed the flavour selection and the final measurement.

### 3.4.1 Extraction

The first stage of this work's pipeline is to convert the released MiniAOD and MiniAODSIM datasets into compact ROOT ntuples with all the extracted information necessary for the rest of the analysis. This stage runs inside the CMS open data container, where the collections present inside the datasets are accessed through FWLite [32], a lightweight Python interface to the CMS event data model. The script developed by this work is capable of analysing datasets without downloading them. In order to archieve that, the dataset's files are listed in the file indices provided with each record on the CERN Open Data Portal and can be read remotelly from the portal's storage. This completely avoids downloading transferring the several terabytes of the full dataset but sacrifices a little bit of the execution time of the script. Since there are multiple files and all different from each other, this script is parallelised as an array of jobs on a computing cluster. Each one of this jobs processes a portion of all the dataset's files, creating for each file its own output file. After all the jobs are completed, a script was made to merge all the output files into a single one with all the extracted variables of the dataset.

For the collision data, each event goes through a sequence of requirements. As required by CMS, each event is first checked against the list of certified luminosity sections, also provided with the datasets, and discarded if its run or luminosity sections are not certified for physics. Then, the event is required to have triggered at least one of the `HLT_AK8PFJet` trigger paths. After passing these two conditions, the jets of the `slimmedJetsAK8` collection are read off and those satisfying the transverse-momentum and pseudorapidity requirements of Table 3.4 are extracted. For each extracted jet, its kinematics are stored together with the list of its constituents, with $p_{\mathrm{T}} > 1$GeV, accompanied by their kinematics and electric charge. Finally, the event is kept only if it contains at least two extracted jets separated in azimuth by more than $2$ radians, in which case it is written to the output tree.

The simulated sample is processed in a similar way with two differences. The run and luminosity certification and trigger requirements are not applied, since they concern real conditions of data taking and have no counterpart in simulation. Because the selected jets lie in the region where the trigger is almost fully efficient, omitting it in simulation does not introduce a difference between the two samples. In addition, a reconstructed jet and its constituents are only extracted if it has an associated generator-level jet, so that every simulated jet entering the analysis has a counterpart at generator level. Furthermore, for each event, the final state partons are also stored. These partons are quarks and gluons of the generator level record that do not split into other partons. In order words, they are the last partons of the shower before hadronisation. These final state partons are stored for the event as a whole, with their kinematics and identity, and are not associated to any reconstructed jet. This step is only done at a later stage described in Section 3.4.3.

The output of this stage is then a ROOT tree per sample, in which each entry corresponds to an event and holds the selected jets, their constituents and, in simulation, the generated partons, as listed in Table 3.3. After adquiring this tree, the CMS software is no longer needed for the remaining stages of this work's pipeline, since it is not in a MiniAOD data format and instead a simple ROOT tree.

### 3.4.2 Declustering and Lund observables

The second stage of this pipeline picks up the output ROOT tree, from the previous stage, and computes, for each jet present, the Lund declustering and the observables built from it. This script is written in C++ and uses ROOT and FastJet [9], running outside the CMS container. Each ntuple file can be processed

independently, so the stage is parallelised across files in the same way as the extraction, although it can also be ran in a single merged output file. The variables computed by this script are stored in the same tree per jet as new branches, so that a single file holds, for every jet, both the extracted information and the observables derived from it.

For each event inside the ROOT tree, the script processes the jets in decreasing order of their transverse momentum, and each jet contributes to the analysis. The stored constituents of the jet are first converted into four-momenta from their transverse momentum, pseudorapidity, azimuth and mass, following equations 3.1-3.2, and each is tagged with its electric charge so that it can be traced back after the clustering process. The constituents are then reclustered with the C/A algorithm of Section 2.2.3, using $R = 2$. This value is far larger than the angular size of an AK8 jet ($0.8$), since it guarantees that all the constituents of a single anti-$k_\mathsf{T}$jet are gathered into a single C/A object and that no part of the jet is lost during the reclustering. Jets with fewer than two constituents cannot be declustered and are flagged as failures.

$$p_{x,i} = p_{\mathsf{T},i} \cos\phi_i, \qquad p_{y,i} = p_{\mathsf{T},i} \sin\phi_i, \qquad p_{z,i} = p_{\mathsf{T},i} \sinh\eta_i, \tag{3.1}$$

$$E_i = \sqrt{p_{x,i}^2 + p_{y,i}^2 + p_{z,i}^2 + m_i^2}. \tag{3.2}$$

After the reclustering, the full history is then declustered with the LundGenerator of the FastJet contrib [1], which follows the harder branch at each step and returns the ordered list of primary declusterings. For each declustering it provides the angular separation $\Delta R$, the relative transverse momentum $k_\mathsf{T}$, the momentum fraction $z$ and the Lund coordinates of Eq. (2.10). Each primary declusterings in order to be recorded has to pass the SoftDrop condition of equation 2.12 with $z_\mathsf{cut} = 0.1$ and $\beta = 0$, reducing it to $z > 0.1$. With this list of primary declusterings, the primary splitting is chosen by selecting the declustering with largest $k_\mathsf{T}$, with its harder and softer branches denoted as $p_1$ and $p_2$. The softer branche $p_2$ is then declustered in the same way, producing a second list of secondary declusterings. These secondary declusterings that satisfy, again, the SoftDrop condition, form the secondary Lund plane, and the one with largest $k_\mathsf{T}$ is selected as the secondary splitting, with harder and softer branches denoted as $p_3$ and $p_4$. If among the list of primary or secondary declusterings noone satisfies the SoftDrop condition, the jet is flagged as a failure and no $\Delta\psi_{12}$ is computed.

Following the selection of the primary and secondary splittings, the azimuthal angle between the two splitting planes is computed from the three-momenta of the four branches, following the convention of the CMS measurement [30]. The normals of the two planes are

$$\hat{n}_1 = \frac{\vec{p}_1 \times \vec{p}_2}{|\vec{p}_1 \times \vec{p}_2|}, \qquad \hat{n}_2 = \frac{\vec{p}_3 \times \vec{p}_4}{|\vec{p}_3 \times \vec{p}_4|}, \tag{3.3}$$

and the angle between them is

$$\Delta\psi_{12} = \arccos(s\,\hat{n}_1 \cdot \hat{n}_2)\,, \qquad s = \mathrm{sign}[(\hat{n}_1 \times \hat{n}_2) \cdot \vec{p}_\mathsf{h}]\,, \tag{3.4}$$

where $\vec{p}_\mathsf{h}$ is the more energetic of $\vec{p}_1$ and $\vec{p}_2$. The sign $s$ orients the two normals consistently, so that $\Delta\psi_{12}$ is defined in the range $[0, \pi]$. Since the two branches of a declustering are always separated in angle, both planes are well defined. Nevertheless, as a safe guard, the script returns no value in the case of exactly collinear branches, for which the cross products of Eq. (3.3) vanish.

To conclude this process, a set of shape variables is computed for each of the four branches $p_1$ to $p_4$ from the constituents it contains. hese are the number of constituents $n_{\text{all}}$, the number of charged constituents $n_{\text{ch}}$, the transverse-momentum-weighted width $\sigma$ and the transverse-momentum-weighted mean constituent momentum $\langle p_{\text{T}} \rangle$, defined as

$$\sigma = \frac{\sum_i p_{\text{T},i}\, \Delta R_i}{\sum_i p_{\text{T},i}}, \qquad \langle p_{\text{T}} \rangle = \frac{\sum_i p_{\text{T},i}^2}{\sum_i p_{\text{T},i}}, \tag{3.5}$$

where the sums run over the constituents of the branch and $\Delta R_i$ is the angular distance of constituent $i$ from the branch axis. These variables play no role in the computation of $\Delta\psi_{12}$. They characterise the internal structure of each branch, which differs between branches initiated by quarks and by gluons, and they serve as the inputs to the flavour selection of Section 3.5. In collision data no truth information is available, so the channel of the second splitting is left undetermined at this stage. The same computation, together with the determination of the splitting channel, is carried out in simulation data and it is described on the following section.

### 3.4.3 Generator-level channel labelling

In simulated data, the reconstructed jets are processed exactly as described in Section 3.4.2, so that the Lund observables obtained are identical to the ones obtained in collision data, however, the channel label of the primary and secondary splittings is also assigned. This label is built from the final-state partons stored during extraction and its construction is made in three steps. Flavoured jets are first built from the partons, these generator-level jets are then matched to the reconstructed jets, and finally the splittings of the two are matched and labelled.

Firstly, the final-state partons of each event are converted into four-momenta in the same way as the reconstructed constituents, through equation (3.1) and (3.2), but, in addition, each parton is also assigned a flavour from their identity. A quark carries one unit of its flavour, an antiquark minus one unit, and a gluon is flavourless. The partons are then clustured using the anti-$k_{\text{T}}$algorithm with $R = 0.8$, mimicking the same definition as the reconstructed jets. Afterwards, it is necessary to match these new generated jets with their respective reconstructed jet. This step is performed in their C/A reclustering forms. On the reconstructed side, this is the object obtained in Section 3.4.2 by reclustering the constituents of the jet with the Cambridge/Aachen algorithm and $R = 2$. On the generator side, the partons of each anti-$k_{\text{T}}$ jet containing at least two partons are reclustered in the same way, with the Cambridge/Aachen algorithm and $R = 2$, combined with the IFN procedure of Section 2.2.4 [13] with its parameter $\alpha$ set to $2$. Each reconstructed jet, taken in order of decreasing transverse momentum, is matched to the closest generator-level object in the $(\eta, \phi)$ plane among those not already matched, provided that

$$\Delta R\big(j_{\text{C/A}}^{\text{reco}},\, j_{\text{C/A}}^{\text{gen}}\big) < 0.4, \tag{3.6}$$

a threshold equal to half the radius of the anti-$k_{\text{T}}$ jets. This matching process is therefore one to one. Since each C/A object contains all the constituents of the anti-$k_{\text{T}}$jet which was built from, its axis should coincide with the axis of the original jet, making it equivalent to matching to matching the anti-$k_{\text{T}}$jets themselves. The generator-level object selected in this way provides a Cambridge/Aachen history in which every branch carries an infrared- and collinear-safe flavour. A reconstructed jet with no generator-level counterpart satisfying Eq. (3.6) receives no truth label and is not used in the training of the flavour

selection.

With the jet matching complete, the generator-level history is then declustered via the same process of as the reconstructed one. he harder branch is followed at each step, each splitting is tested against the condition $z > 0.1$, and among the splittings that satisfy it the one with the largest $k_{\mathrm{T}}$ is taken as the primary generator-level splitting. Its softer branch is declustered in turn to obtain the secondary generator-level splitting. For each of these two splittings, the flavours of their branches are read from the IFN history and the channel of their splitting is labelled according to Table 3.5.

**Table 3.5** – Assignment of the splitting channel from the IFN flavours of the two branches of a generator-level splitting.

| Flavours of the two branches | Channel |
|---|---|
| both flavourless | $g \to gg$ |
| one quark or antiquark and one flavourless | $q \to qg$ |
| a quark and an antiquark of the same flavour | $g \to q\bar{q}$ |
| any other combination | rest |

Note that during this work, the channel of a splitting is denoted as the one inferred from the IFN flavours of its own branches thus, and, although it is an infrared and collinear safe proxy, it may not coincide with coincide with the real truth channel of the splitting. The flavour of a branch reflects the whole of the radiation it contains, and the matching to the reconstructed jet and the hadronisation introduce further differences. Therefore, the modulation obtained for a given label in simulation does not reproduce the theoretical value of [16] for the corresponding channel.

In order to then label the primary and secondary reconstructed splittings, the primary and secondary generator-level splitting have to correspond geometrically to them. For a reconstructed splitting with branches $p_a$ and $p_b$ and opening angle $\Delta_{ab}$, and the generator-level splitting with branches $g_a$ and $g_b$, the requirement is

$$\Delta R(p_a, g_a) < \tfrac{1}{2}\,\Delta_{ab} \quad \text{and} \quad \Delta R(p_b, g_b) < \tfrac{1}{2}\,\Delta_{ab}, \tag{3.7}$$

where the same condition is also accepted with $g_a$ and $g_b$ interchanged. This is applied seperately to the primary splitting, with branches $p_1$ and $p_2$, and to the secondary splitting with branches $p_3$ and $p_4$. When this condition is satisfied, the reconstructed splitting gets labelled with the channel of Table 3.5. When the condition fails to give any channel, the splitting is labelled as unmatched. An unmatched splitting is one for which the reconstructed declustering does not reproduce the partonic branching, typically because hadronisation or detector effects have altered the clustering sequence. This type of splitting carries no relieable flavour information, so it acts as a background which must be rejected.

Subsequently, each simulated jet carries a label for its first and second splittings, taking one of the values unmatched, $g \to q\bar{q}$, $q \to qg$, $g \to gg$ or rest. The label of the second splitting is the truth information on which the flavour selection of Section 3.5 is trained. It is used only as the target of that selection, while its inputs are the reconstructed quantities of Section 3.4.2, which are available in data as well.

### 3.4.4 The energy correlator

The three-point energy correlator of Section 2.4.4 is computed seperately from the declustering, using directly the jet's constituents stored after extraction. This process does not involve any declustering,

grooming or flavour selection, so that it computes the spin correlation through a different route of the Lund observables. The definition of the three-point energy correlator follows that of Ref.[16], based on the original proposal of Ref.[17], with the particle energies replaced by transverse momenta and the opening angles angles between particles are replaced by their distances in the $(\eta, \phi)$ plane as appropriate for a hadron collider.

For each jet, their constituents with non-zero transverse momentum are converted into three-momenta through equation (3.1) and jets with fewer than three constituents are skipped. The angular distance between every pair of constituents is computed as

$$\Delta R_{ij} = \sqrt{(\eta_i - \eta_j)^2 + (\phi_i - \phi_j)^2}, \tag{3.8}$$

with the azimuthal difference computed in the range $(-\pi, \pi]$. Again, this distance is invariant under boosts along the beam axis, making it the natural measure of angular seperation at a hadron collider and it is the same measure used to define the jets themselves. An opening angle in three dimensions would instead shrink with the pseudorapidity of the jet, so that a fixed window in that angle would select different regions of the jet for central and forward jets.

This computation then runs over every unordered triplet of constituents. The three distances of a triplet are ordered as $\Delta R_S \leq \Delta R_M \leq \Delta R_L$, and the triplet is retained only if

$$\sqrt{0.1} < \Delta R_L < 1 \qquad \text{and} \qquad 0.01 < \Delta R_S < 0.1. \tag{3.9}$$

These are the values used for the opening angles in Ref. [**Karlberg:2021kwr**], but for the $(\eta, \phi)$ plane. These two definitions coincide at small angles for central jets, and jets studied here in this work have transverse momenta comparable to the hard scale for which the window was chosen. The window selects configurations with a strong hierarchy of angles. Two of the constituents lie close together and settle a narrow splitting, while the third lies at a much wider angle. This configuration then configuration then approximates two successive splittings with strongle ordered opening angles which is the regime where the correlator is sensitive to the spin of the intermediate gluon.

For any triplet, let $a$ and $b$ be the two constituents seperated by the smallest distance $\Delta R_S$ and $c$ the third. The azimuthal angle $\psi$ is the angle between the plane containing $\vec{p}_a$ and $\vec{p}_b$ and the plane containing their sum and $\vec{p}_c$. It is computed from the three-momenta of Eq. (3.1), through the normals

$$\hat{n}_1 = \frac{\vec{p}_a \times \vec{p}_b}{|\vec{p}_a \times \vec{p}_b|}, \qquad \hat{n}_2 = \frac{(\vec{p}_a + \vec{p}_b) \times \vec{p}_c}{|(\vec{p}_a + \vec{p}_b) \times \vec{p}_c|}, \tag{3.10}$$

as

$$\psi = \text{atan2}(\hat{u} \cdot (\hat{n}_1 \times \hat{n}_2),\ \hat{n}_1 \cdot \hat{n}_2)\,, \qquad \hat{u} = \frac{\vec{p}_a + \vec{p}_b}{|\vec{p}_a + \vec{p}_b|}, \tag{3.11}$$

which returns a signed angle in the range $(-\pi, \pi]$. Each triplet enters the distribution of $\psi$ with the weight

$$w_{abc} = \frac{p_{\mathrm{T},a}\, p_{\mathrm{T},b}\, p_{\mathrm{T},c}}{p^3_{\mathrm{T,jet}}}, \tag{3.12}$$

in accordance with Eq. (2.15). In the script, since the sum runs over ordered triplets, it will differ from a sum over all all orderings of $i$, $j$ and $k$ only by a constant factor. This will affect the normalisation of the distribution, but not its shape and therefore it will not affect the factor $a_2/a_0$. In the simulated datasets, the distribution is also separately for jets in each channel of the second splitting, as labelled in Section 3.4.3, which provides a channel-by-channel reference for the modulation.

The main difficulty of this stage is computation resources, since each jet may have lots of constituents and the number of triplets grows as the cube of their number. To make the computation manageable, the pairwise distances are computed once per jet, the window of Eq. (3.9) is applied before any other quantity is evaluated, so that most triplets are rejected at negligible cost, and the loop over triplets is compiled to machine code with the Numba just-in-time compiler [33].

## 3.5 Flavour selection

The core measurement of this work requires isolating the secondary splitting channel $g \to q\bar{q}$. As discussed in Section 2.4.2, the modulation of this channel carries opposite sign compared to the $g \to gg$, so that an inclusive measurement dilutes the effect, while the other channel $q \to qg$ vanishes. However, the channel of a splitting is only defined in simulation, through generator-level labelling of Section 3.4.3, while in collision data remains undefined.

The approach adopted here in this section is to learn, in simulation, the relation between this true channel and quantities that are measured in the same way in data. For that, a classifier was trained on the reconstructed features of each jet, with the truth label as its target, and is then applied unchanged to the collision data. An important feature is that its output is not a definite label but a score, whose value gives how likely a certain splitting is to belong to the $g \to q\bar{q}$ channel. With that score, one can require it to exceed a threshold and obtain an enriched sample of that channel. The $\Delta\psi_{12}$ observable is then measured as a function of this threshold, with the fraction of $g \to q\bar{q}$ splittings in each selected sample, its purity, determined from simulation.

This section is organised as follows. Section 3.5.1 describes the overall strategy and the classes into which the splittings are divided. Section 3.5.2 presents the input features of the classifier, and Section 3.5.3 the architecture of the networks and the procedure used to train them. Section 3.5.4 describes the alternative approaches that were considered, and Section 3.5.5 explains how the trained networks are applied to the data and how the purity of the selected samples is computed in simulation.

### 3.5.1 Strategy and classes

The generator level-labelling of section 3.4.3 divides the simulated jets into five classes according to their secondary splitting channel. A splitting can either be unmatched, if the reconstructed declustering does not reproduce the partonic branching, or be assigned with the channels $g \to q\bar{q}$, $q \to qg$, $g \to gg$, and rest. The composition of the simulated sample in these classes is given in Table 3.6. As one can notice, the unmatched label represents the majority of the dataset, about three quarters of it, but it does carries no relieable flavour information, while, among the matched splittings, the $g \to q\bar{q}$ channel is a minority, outnumbered by both $q \to qg$ and $g \to gg$.

The selection procedure is therefore divided into two stages, each adressing a part of the dataset. The first stage tries to seperate the matched splittings, of the channels $g \to q\bar{q}$, $q \to qg$ and $g \to gg$, from

**Table 3.6** – Composition of the labelled simulated sample by class of the second splitting, for jets with a valid $\Delta\psi_{12}$.

| Class | Jets | Fraction (%) |
|---|---|---|
| unmatched | 22 689 428 | 75.95 |
| $g \to q\bar{q}$ | 896 936 | 3 |
| $q \to qg$ | 1 736 862 | 5.81 |
| $g \to gg$ | 4 298 461 | 14.39 |
| rest | 253 299 | 0.85 |
| total | 29 874 986 | 100.0 |

the unmatched ones. Its output is a matched score, which is the classifier's confidence level that it is a matched labelled channel. Only the jets with matched score exceeding $0.8$ are passed to the second stage. The second stage is only trained on matched splittings and seperated the $g \to q\bar{q}$ channel from the $q \to qg$ and $g \to gg$ channels taken together. This classifier's output is the $g \to q\bar{q}$ score on which the threshold of the measurement is placed. As suggested in [30], the residual cattegory is excluded from the training of both stages, since it represents less than one percent of the dataset and represents other flavour configurations that do not correspond to any channel.

Dividing the problem in this way makes the two classifiers concentrate on a single binary question. The first stage removes the unmatched background, in which the flavour label is not useful. The second stage then has to solve only the separation that matters for the measurement, without spending capacity on the distinction between $q \to qg$ and $g \to gg$, which plays no role in the selection. The strong imbalance between the classes in both stages is compensated during training, as described in Section 3.5.3.

### 3.5.2 Input features

The inputs of the classifers are built from the primary and secondary splittings of each jet and from the four branches computed in section 3.4.2. They fall into two groups, the shapes of the branches of the second splitting and the kinematics of the two selected splittings, and are listed in Table 3.7.

Each of these variables exploit the different radiation patterns of quark and gluons. A gluon carries the larger colour factor, $C_A = 3$ against $C_F = 4/3$ for a quark, as introduced in Section 2.1, and therefore radiates more. Because of that, a branch initiated by a gluon tends to have more constituents, to be wider, and to share its momentum among softer constituents than a branch initiated by a quark. For each of the two branches $p_3$ and $p_4$ of the secondary splitting, this is captured by the number of constituents $n_{\text{all}}$, the number of charged constituents $n_{\text{ch}}$, the width $\sigma$ and the weighted mean constituent momentum $\langle p_{\text{T}} \rangle$ of equation (3.5). This is supposed to capture the nature of these two branches. A $g \to q\bar{q}$ splitting produces two quark branches, a $g \to gg$ splitting two gluon branches, and a $q \to qg$ splitting one of each.

The kinematic variables exploit the different momentum sharing of the channels. From the splitting functions of equations (2.4) and (2.3), it can be noted that the channels $g \to gg$ and $q \to qg$ have increased probability when one of the two products is soft, while the $g \to q\bar{q}$ channel of equation (2.5), has increased probability with a more even sharing of the momentum. The momentum fraction $z_2$ of the secondary splitting is, therefore, sensitive to its channel and it is shown in section **??** precisely that. It is also complemented by the angular separation $\Delta R_{34}$ and the relative transverse momentum $k_{\text{T},2}$ of the second splitting, and by the momentum fraction $z_1$ of the first.

The first stage uses these twelve variables directly while the second stage is also supplemented by combinations that do not depend on the ordering of the two branches. For each pair of variables and

**Table 3.7** – Input features of the two stages of the flavour selection. The superscripts $(3)$ and $(4)$ refer to the harder and softer branches of the second splitting.

| Feature | Description | Stage 1 | Stage 2 |
|---|---|---|---|
| $z_1$, $z_2$ | momentum fractions of the two splittings | yes | yes |
| $\Delta R_{34}$ | angular separation of the second splitting | yes | yes |
| $k_{\mathrm{T},2}$ | relative transverse momentum of the second splitting | yes | yes |
| $n_{\mathrm{all}}^{(3)}$, $n_{\mathrm{all}}^{(4)}$ | number of constituents | yes | yes |
| $n_{\mathrm{ch}}^{(3)}$, $n_{\mathrm{ch}}^{(4)}$ | number of charged constituents | yes | yes |
| $\sigma^{(3)}$, $\sigma^{(4)}$ | width | yes | yes |
| $\langle p_{\mathrm{T}} \rangle^{(3)}$, $\langle p_{\mathrm{T}} \rangle^{(4)}$ | weighted mean constituent momentum | yes | yes |
| symmetric combinations | sum, difference, minimum, maximum and ratio | no | yes |
| total | | 12 | 37 |

for he pair $z_1$ and $z_2$, the sum the absolute difference, the minimum, the maximum and the ratio of the minimum to the maximum are computed and added to the second classifier. This is motivated by the symmetry of the problem. Since in a $g \to q\bar{q}$ splitting both branches are quarks and in a $q \to qg$ splitting the quark is either the harder or the softer branch, this should help the classifier see the flavour information as a whole and not by its ordering. In the second stage the variables $\langle p_{\mathrm{T}} \rangle$ and $k_{\mathrm{T},2}$, whose distributions have long tails, are also replaced by their logarithms, $\ln(1 + x)$. In both stages every feature is finally standardised to zero mean and unit variance, using the statistics of the training sample.

Moreover, to avoid biasing the measurement, two quantities available for each jet are excluded from the inputs. The first is the observable itself, $\Delta\psi_{12}$. A classifier with access to it could select jets according to their azimuthal angle, and would thereby shape the very distribution that this work measures. The second is the transverse momentum of the jet. The flavour composition of the sample changes with the jet transverse momentum, and the selection is trained and calibrated in different intervals of $\hat{p}_{\mathrm{T}}$. Excluding the jet transverse momentum prevents the classifier from learning this composition, and keeps its output tied to the internal structure of the splitting rather than to the kinematics of the jet. It also allows the same classifier to be applied, in future work, to jets in other ranges of transverse momentum.

### 3.5.3 Network architecture and training

The two stages of the selection are feedforward neural networks, implemented with Keras [34] on top of TensorFlow [35], with the preprocessing of the inputs performed with scikit-learn [36]. The networks share the same structure with three hidden layers of $256$, $128$ and $64$ neurons with rectified linear activations, and is regularised through a penalty on the squared magnitude of its weights, batch normalisation and dropout. Their differences are the strength of the regularisation, in their output layer and in the loss function minimised during training. The complete configuration of the two networks is given in Table 3.8.

The first stage neural network has two output neurons with a softmax activation, thus returning the probabilities that the secondary splitting is matched or unmatched, and the first one is the matched score used in this selection. This network was trained by minimising the categorical cross-entropy and, since the matched splittings make about three quarters of the dataset, each class is weighted in the loss by a factor inversely proportional to its frequency. These weights are softened by raising them to the power $0.75$, which prevents the network from being dominated by the majority class without overcompensating for it.

The second stage has a single output neuron with a sigmoid activation, whose value is the $g \to q\bar{q}$ score.

**Table 3.8** – Architecture and training configuration of the two stages of the flavour selection.

| | Stage 1 | Stage 2 |
|---|---|---|
| Hidden layers | $256$, $128$, $64$ | $256$, $128$, $64$ |
| Activation | ReLU | ReLU |
| L2 penalty | $10^{-4}$ | $10^{-5}$ |
| Batch normalisation | after every layer | after every layer |
| Dropout | $0.25$, $0.20$, $0.10$ | $0.20$, $0.20$, $0.10$ |
| Output | two neurons, softmax | one neuron, sigmoid |
| Loss | categorical cross-entropy | binary focal loss |
| Class weights | balanced, power $0.75$ | balanced |
| Optimiser | Adam, rate $10^{-3}$ | Adam, rate $10^{-3}$ |
| Batch size | $1024$ | $1024$ |
| Maximum epochs | $100$ | $300$ |
| Monitored metric | macro-averaged F1 score | area under the PR curve |

It is trained with the binary focal loss [37],

$$\mathcal{L}_{\text{focal}} = -\,\alpha_t\,(1-p_t)^{\gamma}\ln p_t, \tag{3.13}$$

where $p_t$ is the probability that the network assigns to the correct class, and $\alpha_t$ equals $\alpha$ for splittings of the $g \to q\bar{q}$ channel and $1-\alpha$ for the others. The values $\gamma = 2$ and $\alpha = 0.5$ are used. The factor $(1-p_t)^{\gamma}$ reduces the contribution of splittings that are already classified with confidence, and concentrates the training on the ambiguous ones. This is very fitting to the problem since the channels overlap each other in the space of the input features and most splittings are hard to classify. Balanced class weights are applied furthermore, to compensate for the minority of the $g \to q\bar{q}$ channel among the other splittings.

The difference the two networks is due to the different nature of their task. The first stage network seperates classes which differ more clearly in the kinematics of the splittings, since unmatched splittings are those in which the reconstructed branching departs from the partonic one, and the separation is learned with relative ease. Therefore, this model has a stronger regularisation, with larger weight penalty and more dropout in its first layer, preventing it from learning accidental patterns or noise that exists only in the training dataset and encouraging it to more general patterns that are more likely to work on general unseen data. Meanwhile, the second stage must instead extract the smaller differences between the shapes of quark and gluon branches. It uses lighter regularisation and a longer maximum training time so it can learn these weaker features, while early stopping prevents overtraining in both cases. The class imbalance treatment also differs. In the first stage, neither class is of particular interest, and the softened weights keep the matched score closer to a probability, so that the fixed requirement of $0.8$ retains its meaning. In the second stage the underrepresented group is the one of interest, and fully balanced weights ensure that it is not neglected. Finally, the metric used to select the best epoch matches how each output is used. The macro-averaged F1 score weights the two classes of the first stage equally, while the accuracy of the second stage isprecisely the purity of the selected sample, so that the area under the precision–recall curve measures directly the purity achievable at each efficiency.

Both networks are trained on the simulated dataset with $800 < \hat{p}_{\mathrm{T}} < 1000$ GeV, using the jets whose secondary splitting have been labelled, but excluding the residual cattegory. The first stage network was trained on all such jets while the second stage network only on the matched ones. The sample is divided into training, validation and test subsets containing about two thirds, one sixth and one sixth of the events. The division was made by event, so that the jets of a given event never appear in different subsets.

This step was also tested with a division made by jet and the performance was essentially the same, which was expected since the jets supposably originate from independent showers. The preprocessing of Section 3.5.2, is determined on the training subset alone and then applied unchanged to the other subsets and to the data.

The networks are optimised with the Adam algorithm [38] in batches of $1024$ jets. After each epoch, the performance is evaluated on the validation subset, using the macro-averaged F1 score for the first stage and the area under the precision-recall curve for the second. The latter is better suited than the area under the Receiver Operating Characteristic (ROC) curve to a problem in which the class of interest is a minority. The learning rate is halved whenever this metric fails to improve for five consecutive epochs, down to a minimum of $10^{-6}$, and training stops after fifteen epochs without improvement, at which point the weights of the best epoch are restored. The test subset is used only for the final evaluation of the networks.

### 3.5.4 Alternative approaches

The double stage selection described in the previous sections was used after comparison with three alternative strategies. All these different approaches were built on the same input features and the same preprocessing, and, again, they all exclude the $\Delta\psi_{12}$ variable from their inputs.

The first alternative is a single neural network that classifies each jet into the four categories unmatched, $g \to q\bar{q}$, $q \to qg$ and $g \to gg$. It's structure is the same as of the double stage selection with the same hidden layers described in Table 3.8, a softmax output with four neurons, and is trained with the categorical cross-entropy and class weights softened with the power $0.75$. This strategy attempts to solve the two problems that the two stage neural networks approach individually. A network of this kind is used by CMS in [30], where it was trained on a different generator and it tries to reject the unmatched splittings and identify the $g \to q\bar{q}$ channel.

The second approach replaces the binary second stage by a network with three outputs, trained on the matched splittings to separate $g \to q\bar{q}$, $q \to qg$ and $g \to gg$, with the same loss and class weights as the first alternative. It is applied after the same first stage, and its $g \to q\bar{q}$ output is used for the selection.

The third alternative is an unsupervised neural network. The matched splittings are described by the same features, reduced with a principal component analysis, and a Gaussian mixture model with full covariance matrices is fitted to them for several numbers of components, without any use of the flavour labels. The resulting clusters are then compared with the channel composition, in order to test whether the structure of the feature space separates the channels by itself.

Although these approaches seem fair, none of them could seperate the $g \to q\bar{q}$ channel as effectively as the two-stage selection, as shown in Chapter 4. The reason behind this may be because the $g \to q\bar{q}$ channel is a minority of the sample, and a network that must also seperate the channels $q \to qg$ and $g \to gg$, or reject the unmatched splittings at the same time, spends part of its power on that which it is not required on this measurement. In the unsupervised case, the channels overlap strongly in the space of the input features, so that there is no reason for the clusters of a Gaussian mixture to align with them, and the clusters found indeed mix the channels. The two-stage selection is therefore the one used in the remainder of this work.

### 3.5.5 Application and purity calibration

After training, the two stage networks are applied unchanged to every jet with a valid $\Delta\psi_{12}$, both in the collision data and in the simulated calibration sample. The inputs of each jet are built and preprocessed exactly as in training, using the same engineered features and the same imputation and standardisation, determined on the training subset. Thus, each jet receives two scores, the matched score of the first stage and the $g \to q\bar{q}$ score of the second, which are then stored alongside the other jet quantities.

Afterwards, the selection is made by picking jets with a matched score above $0.8$, and a $g \to q\bar{q}$ score above a threshold $t$. The first requirement is fixed and removes most of the unmatched splittings. The threshold $t$ varies, and each of its values defines a selected sample. As $t$ increases, the samples become smaller and progressively enriched in the $g \to q\bar{q}$ channel.

For each threshold $t$, the composition of the selected sample is computed on the simulated sample with $600 < \hat{p}_{\mathrm{T}} < 800$ GeV, which is independent of the one used in training. The purity and the efficiency of the selection are defined as

$$P(t) = \frac{N_{q\bar{q}}(t)}{N(t)}, \qquad \varepsilon(t) = \frac{N_{q\bar{q}}(t)}{N_{q\bar{q}}}, \tag{3.14}$$

where $N(t)$ is the number of selected jets, $N_{q\bar{q}}(t)$ the number of them whose secondary splitting is labelled $g \to q\bar{q}$, and $N_{q\bar{q}}$ the total number of jets carrying that label before the selection. When increasing the threshold, the purity grows while the efficiency declines, and the comparison between the two is shown in Chapter 4 summarising the performance of the selection.

At each threshold, the mean value $\langle \cos 2\Delta\psi_{12} \rangle$ of equaiton (3.15) is computed on the selected jets, seperately in data and simulation, with the statistical uncertainty of equation (3.16). The measurement made is presented as a function of the threshold, with the purity of equation (3.14) given for each value. The two reference are computed from simulation and data are the mean over all jets labelled $g \to q\bar{q}$ and the mean over those labelled $q \to qg$ or $g \to gg$, which indicates the value expected for the labelled channel and for the background before any selection. The evaluated threshold range is from $0.3$ up to $0.6$ and beyond this value the number of jets fall below a few hundred and the relative statistical uncertainty exceeds $10$%, making these points carry little information.

The main assumption of this procedure is that the purity computed in simulation samples also describes the samples selected from data at the same threshold. The networks are identical in the two cases, but the composition of the selected samples depends on how well the simulation reproduces the data, as discussed in Section 3.6.2.

## 3.6 Systematic uncertainties

This work's quantity is the mean value $\langle \cos 2\Delta\psi_{12} \rangle$, given per threshold of the $g \to q\bar{q}$ score organized in samples, accompanied by the purity assigned to each sample, according to simulation. The $\langle \cos 2\Delta\psi_{12} \rangle$ and purity quantities are both subject of uncertanties. The statistical uncertanty comes from the finite number of jets in each sample and it is evaluated at every point of this measurement. The systematic uncertanties arise from the modelling of the simulation on which the selection is trained and calibrated, from the response of the detector, and from some choices made in this analysis. However, unlike the

statistical uncertainties, they do not decrease when extracting more jets and they may shift the result in a definite result

A complete quantitive evaluation of every systematic uncertanty was not done in this work. This section just indentifies the relevant souces to the measurement, describes how each of them is managed by this analysis, and states which ones are not evaluated. Section 3.6.1 describes the statistical uncertainty. Section 3.6.2 discusses the modelling of the purity, which is the dominant systematic source. Section 3.6.3 addresses the effects of the detector, and Section 3.6.4 the trigger and the choices of the analysis. An overview of the sources considered and of the treatment of each is given in Table 3.9, and each source is discussed in turn in the subsections that follow.

**Table 3.9** – Overview of the sources of uncertainty and of their treatment in this work. Each source is discussed in the section indicated.

| Source | Treatment | Section |
|---|---|---|
| Statistical | Evaluated for every selected sample | 3.6.1 |
| Modelling of the purity | Not evaluated, single generator | 3.6.2 |
| Kinematic dependence of purity | Computed on an independent sample | 3.6.2 |
| Detector resolution | Present in data and simulation, not unfolded | 3.6.3 |
| Jet energy scale | Expected small, not evaluated | 3.6.3 |
| Pile-up | Suppressed by $z > 0.1$, overlaid in simulation | 3.6.3 |
| Trigger | Negligible above $700$ GeV | 3.6.4 |
| Analysis choices | Fixed, not varied | 3.6.4 |

### 3.6.1 Statistical uncertainty

For a selected sample of $N$ jets, the measured quantity is the mean computed as

$$\langle \cos 2\Delta\psi_{12} \rangle = \frac{1}{N} \sum_{j=1}^{N} \cos 2\Delta\psi_{12}^{(j)}, \tag{3.15}$$

and its statistical uncertainty is the standard error of the mean,

$$\delta_{\text{stat}} = \frac{s}{\sqrt{N}}, \qquad s^2 = \frac{1}{N-1} \sum_{j=1}^{N} \left( \cos 2\Delta\psi_{12}^{(j)} - \langle \cos 2\Delta\psi_{12} \rangle \right)^2 . \tag{3.16}$$

For a distribution of the form of equation (2.14), the mean value is $\langle \cos 2\Delta\psi_{12} \rangle = \frac{1}{2}\, a_2/a_0$, so that the uncertainty on $a_2/a_0$ is twice that of Eq. (3.16).

This expression assumes that each jet entering the sample is independent. In each event after selection, there is at least two jets and all of them enter the analysis, although, events with more than two jets are rare. In that case, the jets of an event originate from separate showers, and their second splittings are essentially uncorrelated. This is statement is supported by the behaviour of the flavour selection, because its performance did not change, when training and test samples were separated by event rather than by jet.

The purity given for each sample is the fraction of selected simulated jets whose second secondary splitting is labelled $g \to q\bar{q}$ by the IFN procedure. Its statistical uncertainty follows from the binomial distribution,

$$\delta P = \sqrt{\frac{P\,(1-P)}{N_{\text{MC}}}}, \tag{3.17}$$

where $N_{\text{MC}}$ is the number of selected simulated jets. The same expression of equation (3.16) gives the statistical uncertainty of the mean values obtained in simulation. In this work, when showing the results as a function of the threshold, the threshold itself is an exact choice, given by the classifiers, and carries no uncertanty, so that only the vertical uncertainties of equation (3.16) are presented.

There are two features of the statistical uncertainties that should be kept in mind while reading the results. The first is that it grows with the threshold, since raising it catches fewer jets, making the threshold's range limited by the number of jets that survive. The second is that each sample is a subset of the samples selected at lower thresholds, so that consecutive points share most of their jets and their fluctuations are correlated. A trend across thresholds should therefore be judged against the uncertainty of the individual points rather than against the smoothness of the curve.

### 3.6.2 Modelling of the purity

As said on the previous subsection, the purity is the fraction of the selected simulated jets whose second secondary splitting is labelled $g \to q\bar{q}$ by the IFN procedure. It is taken from the simulation and transferred to the data, and it is therefore correct for the data only if the simulation reproduces two of their features. The first is the flavour compostion of the secondary splitting, in particular the rate of $g \to q\bar{q}$ splittings relative to $q \to qg$ and $g \to gg$. The second is the distribution of the input features of Table 3.7 within each channel. If one of them is mismodelled, the same threshold selects a sample of different purity in data than in simulation.

There is direct evidence that this effect is not negligible. In the CMS measurement on which this analysis is based [30], neither Pythia 8 nor Herwig 7 describes the distribution of the classifier output in data, owing to the mismodelling of several input variables, and the simulation is reweighted to reproduce the data before the purity is determined, with an associated uncertainty derived from the difference in flavour composition between the two generators. No such reweighting is applied in this work, and the purities quoted are those of Pythia 8 with the CP5 tune. A repetition of the calibration with a generator based on a different parton shower and hadronisation model was not performed, and the modelling of the purity is therefore the largest source of systematic uncertainty that is not quantified here.

The labelling done in Section 3.5.1, provides a partial consistency check. The composition showed in Table 3.6 agrees closely with the one obtained in [30]. This confirms that the labels are constructed correctly, but it does not prove that the simulation describes in the data.

Another aspect concerning the purity is its depence on the kinematics of the jets. The selection is trained on simulation with $800 < \hat{p}_{\text{T}} < 1000$ GeV and computed on the independent interval $600 < \hat{p}_{\text{T}} < 800$ GeV, which secures that the purity is not biased by the training. Since $\hat{p}_{\text{T}}$ is the transverse momentum of the partonic hard process, each sample contains jets over a brader range of transverse momenta than its nominal interval, and both jets above the selection threshold of $700$ GeV. The distribution of jet transverse momenta in a single interval of $\hat{p}_{\text{T}}$ but still differs from the data, which receives contributions from all intervals, weighted by their cross sections. Although the jet transverse momentum is not a variable which the classifier uses, the relation between threshold and purity may still vary with it, so that a calibration performed on a single interval may not describe the full data sample exactly. Combining several intervals,

each weighted by its cross section, would reproduce the spectrum of the data and remove this limitation.

In addition, the purity depends on the definition of the flavour label. In this work, it is defined as the matching of the branches of the second splitting, $p_3$ and $p_4$, to their generator-level counterparts. However, other choices are possible and, for example, the purities quoted in [30] are obtained by matching the branches $p_2$ and $p_3$ instead, yielding far fewer unmatched splittings, thus a different result. Purities obtained with different definitions are, of course, not directly comparable.

### 3.6.3 Detector effects

This measurement is performed on reconstructed particles which are not corrected to the level of stable particles, making every quantity affected by the finite resolution of the detector. The most relevant effect for an angular observable is the resolution on the directions of the constituents. For charged particles, the direction is measured by the inner tracker with high precision. For neutral particles, it is determined only by the calorimeters, whose granularity in the barrel corresponds to cells of about $0.0174 \times 0.0174$ in $(\eta, \phi)$ for the electromagnetic calorimeter and $0.087 \times 0.087$ for the hadron calorimeter [22]. Splittings resolved at angles comparable to these sizes, are thus measured less precisely which is particularly relevant for the lower edge $\Delta R_S > 0.01$ of the window of the energy correlator. The smearing of the directions blurs the orientation of the reconstructed planes and dilutes the modulation, so that the measured values of $\langle \cos 2\Delta\psi_{12} \rangle$ are expected to be somewhat smaller in magnitude than at particle level.

The simulation dataset is done with a full simulation of the CMS detector, so that data and simulation are compared at the same level and this dilution is present in both. Therefore, comparing both the datasets is consistent, however the values obtained are specific to the CMS detector and cannot be compared with calculations at particle level. Such a comparison would require an unfolding of the detector response, as performed in [30], which is not done in this work.

The jets come with the standard energy corrections done by CMS, whose residual uncertainty moves jets across the selection threshold of $700$ GeV. Since the measured observable is an angle averaged by many jets, and the jets near the threshold form a small part of the sample, this effect is expected to be small, and it is not evaluated.

Some particles comming from other proton–proton interactions in the same bunch crossing, also known as pile-up, enter the jets as soft constituents spread over wide angles. CMS applies a specific pile-up mitigation algorithm in [30], in order to remove this effect. In this work the same method was not applied and just the condition of $z > 0.1$ was imposed on the selected splittings, which rejects the soft branches in which pile-up particles are mostly found. Once again, the simulated samples include an overlay of pile-up interactions, so that its effect is present in data and simulation.

### 3.6.4 Trigger and analysis choices

The trigger is not expected to introduce a significant uncertanty, since, in CMS, a trigger requiring a jet with a transverse momentum above $550$ GeV was found to be more than $99.5$% efficient for jets above $700$ GeV [30]. Thus, the trigger paths used here in this analysis, are expected to have almost full efficiency, but this efficiency is not measured in this work.

Regarding some analysis choices done along the development of this work, several parameters were fixed. These are the grooming condition $z > 0.1$ applied to the selected splittings, the requirement of a

matched score above $0.8$ in the first stage of the selection, the angular window of the energy correlator, and the criteria used to match the generator-level jets and splittings to the reconstructed ones. Playing with these conditions would probe the stability of this work's result, however such thing was not done here.

Of the sources discussed in this section, the two related to the purity are the ones that most limit the interpretation of the result. They are the reason for presenting the measurement as a function of the threshold, with the purity from simulation given as additional information rather than as a calibrated axis. A full evaluation of these sources, based on an alternative generator and on the variations of the analysis choices listed above, is the most important extension of the present analysis and is discussed in Chapter 5.

# 4 RESULTS AND DISCUSSION

# 5 CONCLUSIONS

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